EUROPEAN COMMISSION
DIRECTORATE-GENERAL RESEARCH & INNOVATION
Advanced Manufacturing Systems and Biotechnologies

ANNEX 1 (part A)

Research and Innovation action

NUMBER — 768636 — LASER4SURF

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1.1. The project summary

Project Number [1]	768636	Project Acronym [2]	LASER4SURF

One form per project

General information

Project title [3]	LASER FOR MASS PRODUCTION OF FUNCTIONALISED METALLIC SURFACES
Starting date [4]	01/10/2017
Duration in months [5]	36
Call (part) identifier [6]	H2020-FOF-2017
Topic	FOF-06-2017 New product functionalities through advanced surface manufacturing processes for mass production
Fixed EC Keywords	Surface science and nanostructures, Surface modification
Free keywords	LIPPS, laser, femtoseconds, functionalisation,

Abstract [7]

The main goal of the LASER4SURF project is to develop a laser based solution for the functionalization of metallic surfaces with textures ~1µm or less on mass production. This solution will be based on surface functionalization with ultrashort pulses via Laser Induced Periodic Surface Structures (LIPSS).

The LASER4SURF project will develop a Laser Texturing Prototype to overcome the current barriers for the LIPSS technique to reach mass production, achieving a production rate of 0.1 min/cm2:

- High process time for large 3D pieces.
- Need of inline inspection methods.
- Need of modelling tools for parameter selection.

The development will be focused on the three main steps of the Laser texturing value chain:

- Laser technology/equipment. To generate LIPSS of different pattern sizes with multiple orientations, LASER4SURFwill develop an easy to integrate and control, compact and low cost optical module.
- In-line inspection tool. A whole optical measurement system based on diffractometry technology will be integratewith the laser equipment including hardware and software in order to be able to monitor all different nano properties from the functionalised surfaces.
- Simulation tool that will show automatically the process parameters required to obtain the desired functionalityfor a specific material. The simulation tool will automatically transfer the optimum laser parameters to the LIPSS equipment in order to configure it for the desired pattern.

The project will combine the developments in an all in one solution, that will be validated by three different use case products that represent a broad range of the main industries, metal alloys used and required functionalities. The LASER4SURF project will be developed by a well-balanced consortium that brings together 8 partners: 3 technology developers (CEIT, MULTITELL and VISUM), an integrator of the solutions (LASEA), three industrial partners (FAGOR, RESCOLL and CIC) and a dissemination partner (ESCI)

1.2. List of Beneficiaries

Project Number [1]	768636	Project Acronym [2]	LASER4SURF

List of Beneficiaries

No	Name	Short name	Country	Project entry month[8]	Project exit month
1	ASOCIACION CENTRO TECNOLOGICO CEIT-IK4	Ceit-IK4	Spain	1	36
2	LASER ENGINEERING APPLICATIONS SA	LASEA	Belgium	1	36
3	MULTITEL	MULTITEL	Belgium	1	36
4	IRIS ADVANCED ENGINEERING LIMITED	VISUM	Ireland	1	36
5	CENTRO DE INVESTIGACION COOPERATIVADE ENERGIAS ALTERNATIVAS FUNDACION	CICe	Spain	1	36
6	FAGOR AUTOMATION S COOP LTDA	FAGOR	Spain	1	36
7	RESCOLL	RESCOLL	France	1	36
8	EUROPEAN SCIENCE COMMUNICATION INSTITUTE (ESCI) GGMBH	ESCI	Germany	1	36

1.3. Workplan Tables - Detailed implementation

1.3.1. WT1 List of work packages

WP Number[9]	WP Title	Lead beneficiary[10]	Personmonths[11]	Start month[12]	End month[13]
WP1	PROJECT COORDINATION AND MANAGEMENT	1 - Ceit-IK4		1	36
WP2	DESIGN AND DEVELOPMENT OF A PROTOTYPE FOR CONTROLLED LIPSS	3 - MULTITEL		1	24
WP3	DESIGN AND DEVELOPMENT OF AN IN-LINE MONITORING UNIT	4 - VISUM		1	24
WP4	MODELLING FOR TOOLS FOR LIPSS PROCESES AND PARAMETER SELECTION	1 - Ceit-IK4		1	24
WP5	INTEGRATION OF THE DEVELOPED TOOLS	2 - LASEA		1	33
WP6	TECHNOLOGY VALIDATION IN THREE FIELDS	6 - FAGOR		18	36
WP7	GATHERING EXPERIENCE AND NEW SCENARIO DEFINITION	1 - Ceit-IK4		25	36
WP8	DISSEMINATION AND EXPLOITATION	2 - LASEA		1	36
WP9	Ethics requirements	1 - Ceit-IK4		1	36
		Total			

1.3.2. WT2 list of deliverables

Deliverable Number[14]	Deliverable Title	WP number[9]	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D1.1	Project Management Handbook	WP1	1 - Ceit-IK4	Report	Confidential, only for members of the consortium (including the Commission Services)	3
D1.2	Data Management Plan.	WP1	1 - Ceit-IK4	ORDP: Open Research Data Pilot	Confidential, only for members of the consortium (including the Commission	6

					Services)	
D1.3	IPR Management Plan	WP1	1 - Ceit-IK4	Report	Confidential, only for members of the consortium (including the Commission Services)	6
D2.1	Report on detailed specifications from all the applications	WP2	2 - LASEA	Report	Confidential, only for members of the consortium (including the Commission Services)	6
D2.2	Report on optical configuration of the optical module(s)	WP2	3 - MULTITEL	Report	Confidential, only for members of the consortium (including the Commission Services)	15
D2.3	Report on definition of the interfaces of the optical module	WP2	2 - LASEA	Report	Confidential, only for members of the consortium (including the Commission Services)	18
D2.4	Optical modules prototypes to be integrated in WP5	WP2	3 - MULTITEL	Demonstrator	Confidential, only for members of the consortium (including the Commission Services)	24
D3.1	Design and delivery of the system for lab testing (hardware and software)	WP3	4 - VISUM	Demonstrator	Confidential, only for members of the consortium (including the Commission Services)	7

Deliverable Number[14]	**Deliverable Title**	**WP number[9]**	**Lead beneficiary**	**Type[15]**	**Dissemination level[16]**	**Due Date (in months)[17]**

D3.2	User Manual of the monitoring system	WP3	4 - VISUM	Report	Confidential, only for members of the consortium (including the Commission Services)	24
D3.3	In-line monitoring module prototype to be integrated in WP5	WP3	4 - VISUM	Demonstrator	Confidential, only for members of the consortium (including the Commission Services)	24
D4.1	Database of LIPSS processes and related functionalities	WP4	1 - Ceit-IK4	Other	Confidential, only for members of the consortium (including the Commission Services)	24
D4.2	Report on process parameter determination	WP4	1 - Ceit-IK4	Report	Confidential, only for members of the consortium (including the Commission Services)	24
D4.3	Preliminary Software platform for parameter selection	WP4	1 - Ceit-IK4	Other	Confidential, only for members of the consortium (including the Commission Services)	22
D4.4	Optimized and validated software platform	WP4	1 - Ceit-IK4	Other	Confidential, only for members of the consortium (including the Commission Services)	24
D5.1	Definition of setup specifications of the modules in the global prototype	WP5	2 - LASEA	Report	Confidential, only for members of the consortium (including the Commission Services)	12
D5.2	Integration of the three modules in an already existing working station	WP5	2 - LASEA	Demonstrator	Confidential, only for members of the consortium (including the Commission Services)	24

D5.3	Report on Tests and validation	WP5	2 - LASEA	Report	Confidential, only for members	33

Deliverable Number[14]	Deliverable Title	WP number[9]	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
					of the consortium (including the Commission Services)	
D5.4	Public report Conclusions on LASER4SURF system validation	WP5	2 - LASEA	Report	Public	33
D6.1	Current collector functionalized and validated	WP6	5 - CICe	Demonstrator	Confidential, only for members of the consortium (including the Commission Services)	36
D6.2	Dental implant functionalized and validated	WP6	7 - RESCOLL	Demonstrator	Confidential, only for members of the consortium (including the Commission Services)	36
D6.3	Optical encoder functionalized and validated	WP6	6 - FAGOR	Demonstrator	Confidential, only for members of the consortium (including the Commission Services)	36
D6.4	Summary report on test performed on batteries	WP6	5 - CICe	Report	Confidential, only for members of the consortium (including the Commission Services)	36
D6.5	Summary report on test performed on implants	WP6	7 - RESCOLL	Report	Confidential, only for members of the consortium (including the Commission Services)	36
D6.6	Summary report on test performed on encoders	WP6	6 - FAGOR	Report	Confidential, only for members of the consortium (including the Commission Services)	36

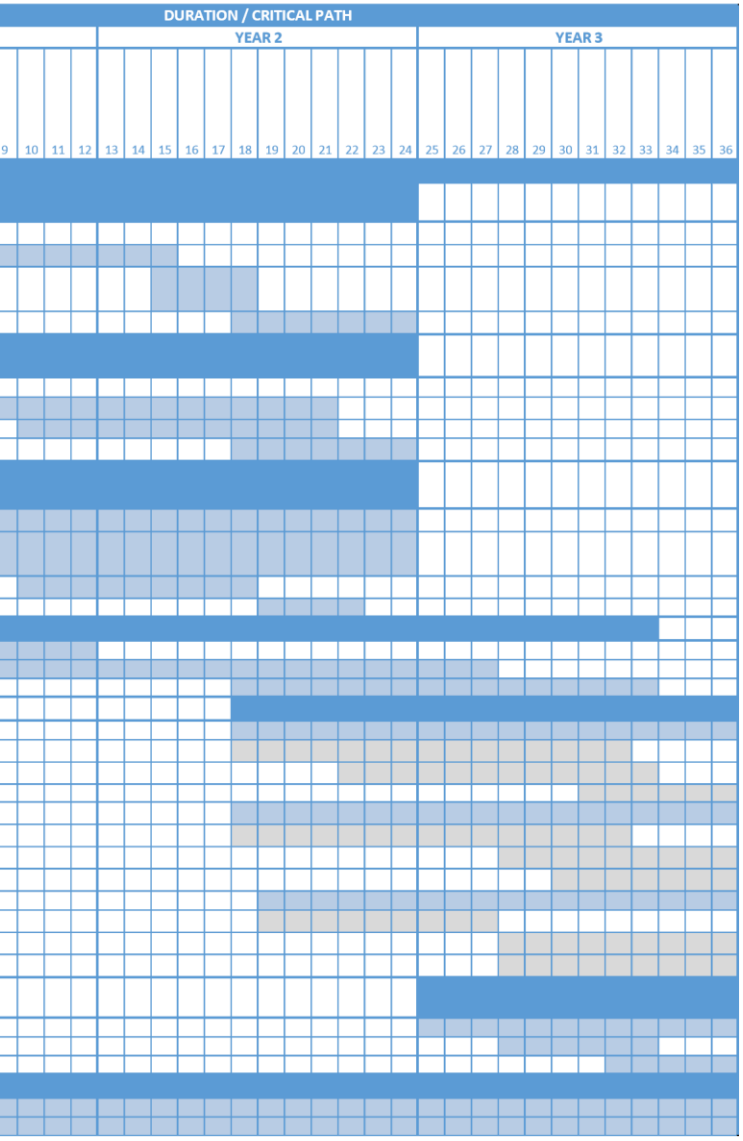
D7.1	Benchmark tables and analysis of the use cases	WP7	1 - Ceit-IK4	Report	Public	33
D7.2	List of potential replications sectors and new applications	WP7	1 - Ceit-IK4	Report	Public	36

Deliverable Number[14]	**Deliverable Title**	**WP number[9]**	**Lead beneficiary**	**Type[15]**	**Dissemination level[16]**	**Due Date (in months)[17]**
D8.1	Business models and exploitation plans	WP8	2 - LASEA	Report	Confidential, only for members of the consortium (including the Commission Services)	12
D8.2	IPR management	WP8	1 - Ceit-IK4	Report	Confidential, only for members of the consortium (including the Commission Services)	36
D8.3	Communication Master Plan	WP8	8 - ESCI	Report	Confidential, only for members of the consortium (including the Commission Services)	3
D8.4	Project website and social media channels	WP8	8 - ESCI	Websites, patents filling, etc.	Public	4
D8.5	Introductory project video	WP8	8 - ESCI	Websites, patents filling, etc.	Public	4
D8.6	Brochure	WP8	8 - ESCI	Websites, patents filling, etc.	Public	6
D8.7	Final project video	WP8	8 - ESCI	Websites, patents filling, etc.	Public	36
D8.8	Interim report on dissemination and communication activities	WP8	8 - ESCI	Report	Confidential, only for members of the consortium (including the Commission Services)	20

D8.9	Final report on all dissemination and communication activities	WP8	8 - ESCI	Report	Confidential, only for members of the consortium (including the Commission Services)	36
D9.1	EPQ - Requirement No. 1	WP9	1 - Ceit-IK4	Ethics	Confidential, only for members of the consortium (including the Commission Services)	6

Associated with document Ref. Ares(2017)3062700 - 19/06/2017

WORKPACKAGE DESCRIPTIONS	EFFORT (PMs)									
	1. CEIT	2. LASEA	3. MULTITEL	4. VISUM	5. Iris (TP)	6. CiCe	7. Cegasa (TP)	8. FAGOR	9. RESCOLL	10. ESCI
WP1 PROJECT MANAGEMENT	L	P	P		P	P	P	P	P	P
WP2 Design and Development of a Prototype for Controlled LIPSS	P	P	L	P		P		P	P	
Task 2.1 Project Requirement Analysis and Definition of the System Architecture	P	P	L	P		P		P	P	
Task 2.2 Preparation of the experimental laboratory set-up and tests	P	P	L							
Task 2.3 Definition of the optical, electrical and mechanical interfaces of the optical modules	P	L	P							
Task 2.4 Realisation and test of the optical modules	P	P	L							
WP3 DESIGN AND DEVELOPMENT OF AN IN-LINE MONITORING UNIT	P	P		L	P			P	P	
Task 3.1 Core optical system development	P	P		L						
Task 3.2 Data acquisition and processing software development	P			P	L					
Task 3.3 Functional prototype development		P		P	L					
Task 3.4 Nano-metrology lab testing and validation	P	P		L						
WP4 MODELLING FOR TOOLS FOR LIPSS PROCESES AND PARAMETER SELECTION	L	P	P			P		P	P	
Task 4.1 Generation of a database of LIPSS processes and resulting functionalities	L	P	P			P		P	P	
Task 4.2 Statistical analysis of the influence of process parameters on the obtained LIPSS and functionalities	L									
Task 4.3 Design and development of a software platform for parameter selection	L	P	P							
Task 4.4 Pre-validation of the software tool	L	P	P							
WP5 INTEGRATION OF THE DEVELOPED TOOLS	P	L	P	P	P		P	P	P	
Task 5.1 Study of the integration	P	L	P	P	P		P	P	P	
Task 5.2 Integration of the diffrent modules	P	L	P	P	P					
Task 5.3 Test and validation	P	L	P	P	P					
WP6 TECHNOLOGY VALIDATION IN THREE FIELDS	P	P				P	P	L	P	
Task 6.1 Technology Validation in Energy Storage.	P	P				L	P			
Subtask 6.1.1 Fabrication of structured current collectors for pouch cell prototypes	P	P				L	P			
Subtask 6.1.2 Monolayer pouch cell fabrication and electrochemical characterization.	P	P				L	P			
Subtask 6.1.3 Characterisation of pouch cell electrodes	P	P				L	P			
Task 6.2 Technology Validation in Biomedical Implants		P							L	
Subtask 6.2.1 Fabrication and dimensional characterization of modified implants		P							L	
Subtask 6.2.2 Mechanical and chemical analysis and validation of the modified implants									L	
Subtask 6.2.3 Biological validation of the added functionality									L	
Task 6.3 Technology Validation in Linear Encoders	P	P						L		
Subtask 6.3.1 Design and Implementation of a validation setup								L		
Subtask 6.3.2 Fabrication and dimensional/optical characterization of diffraction gratings	P	L						P		
Subtask 6.3.3 *Validation of the fabricated gratings in optical encoders*	P							L		
WP7 GATHERING EXPERIENCE AND NEW SCENARIO DEFINITION	L	P	P			P	P	P	P	
Task 7.1 Use case analysis	L	P	P			P	P	P	P	
Task 7.2 Identification of potential replication sectors	P	L					P	P	P	
Task 7.3 Replication guides	L	P	P	P						
WP8 DISSEMINATION AND EXPLOITATION	P	L	P	P	P	P	P	P	P	P
Task 8.1 Exploitation	P	L	P	P	P	P	P	P	P	
Task 8.2 Communication & Dissemination	P	P	P	P	P	P		P	P	L



1.3.3. WT3 Work package descriptions

Work package number [9]	WP1	**Lead beneficiary** [10]	1 - Ceit-IK4
Work package title	PROJECT COORDINATION AND MANAGEMENT		
Start month	1	**End month**	36

Objectives

The objectives of the WP1 is to manage the overall project with respect to administrative, financial and technical issues to ensure the highest quality of service and completing deliverables according to time frame and foreseen budget. Furthermore, it aims at monitoring and reporting the status of the project to be able to take corrective actions if necessary.

Description of work and role of partners

WP1 - PROJECT COORDINATION AND MANAGEMENT [Months: 1-36]
Ceit-IK4, LASEA, MULTITEL, VISUM, CICe, FAGOR, RESCOLL, ESCI
Task 1.1. Project administration: The administrative and financial management of the project
Lead Partner: CEIT - Task duration: M1-M36

The Project Coordinator, CEIT, will be the liaison between the partners and the European Commission. CEIT will be in charge of the coordination and management of administrative and financial matters. This will involve: (i) Giving administrative and financial support to all partners; (ii) Financial management (collection and processing of the partners' cost statements, the distribution of the individual payments to the partners, the controlling of the overall budget and the settlement of all financial issues with the Commission). (iii) Supervising and informing all participants about the project progress (i.e. sending interim reports, meetings minutes, etc.). (iv) Organising project meetings and calling on participants to attend the project meetings, as well as setting up and circulating the meeting agendas.

During the preliminary phase of the project, CEIT will elaborate a project management handbook for the project management and quality assurance. It will include guidelines for financial reporting, presentation standards for deliverables and reports to the European Commission, measures to ensure timely reporting, payment procedures plan and calendar, etc. Internal communication plan will be defined here with the aim of achieving the best results on the project execution. The document will include an internal review procedure to guarantee the quality of the results of the project. This will create the necessary structure for a successful project management and monitoring. The results of this task will be reported in D1.1.

Outcomes: Project management handbook

Task 1.2. Consortium meetings
Lead Partner: CEIT - Participant Partners: All partners - Task duration: M1-M36

Based on the established management structure for the project the following set of meetings is foreseen:
- Kick-off meeting will be held at the beginning of the project, during the first month, and all project partners will beinvited to participate. The place will be defined by the coordinator.
- Project Management Board meetings will be scheduled once a year, but can be called whenever necessary. Asfurther described in section 3.2, the Project Management Board will be responsible of the major decisions affecting the implementation and success of the project.
- Technical Committee meetings will be schedule every six months to monitor the fulfilment of the work programme,the achieved work progress and the quality of the results obtained by the project during the previous months. Technical Committee meetings will be organized every 6 months with a final meeting at the end of the project. Some of these meetings will be organised at the same time as Project management Board meetings to optimize travels expenses and time.

Outcomes: Meeting minutes. Effective Coordination between partners.

Task 1.3. Project reporting
Lead Partner: CEIT - Participant Partners: All partners - Task duration: M1-M36

This task will be the responsibility of the Project Coordinator and will include all the reporting aspects as described in the Grant Agreement. The objective of this task is to manage three Technical Progress reports and Financial Reports and Financial Reports during the whole project (Month 12, 24 and 36). Interim Activity Reports will be also prepared

in between Progress Reports (month 6, 18, 30) in order to have a continuous follow up of project progress. The Project Coordinator will be in charge of requesting and collating the reports and deliverables and sending them to the European Commission. The results drawn will be discussed in the periodic meetings.

Outcomes: Periodic reports

Task 1.4. Risk Management
Lead Partner: CEIT - Participant Partners: All partners - Task duration: M1-M36

The main sources of risks have already been identified during the project preparation phase and contingency plans have already been established. However, unexpected risks can arise during the project lifetime. The Project Coordinator will be in charge of the overall risk management plan and the Project Management Board will be responsible for proactively managing the risks. This task includes: i) Identification of any risk that may arise during the project implementation; ii) Evaluation of the causes and effects of the identified risks and their relevance for the proper project implementation, iii) Definition of alternative tasks/pathways and iv) Management of the implementation of the alternative tasks/pathways.

Outcomes: Any possible risk is successfully managed.

Task 1.5 Data Management
Lead Partner: CEIT - Participant Partners: All partners - Task duration: M1-M36

LASER4SURF is fully aware with the open access to scientific publications article as stated in the Article 29.2 of the H2020 Grant Agreement Open access to scientific publications. In this sense, all the peer-reviewed publications arising from the project will be made freely and openly available via online repository. The actions which will be taken by the project will be:
- All presentations, contributions and publications even partially funded by the project will include the project logo andprominently acknowledge the grant agreement number.
- The publications funded by the project will be uploaded to some specific Bibliographic social networks in order toguarantee open and free access to them.
- Include the following metadata under a standard format in the publication: Terms: “European Union (EU)" and"Horizon 2020"; Title of the project, acronym and grant agreement number; Publication date and Persistent identifier:
LASER4SURF -xxxxx (x meaning the GA number).

Outcomes: Data Management Plan.

Task 1.6. IPR Management
Lead Partner: CEIT - Participant Partners: All partners - Task duration: M1-M36

Given the clear focus on exploitation of results in this project, the partners will specify, and agree on, clear IPR management rules in a Consortium Agreement to be signed by all partners. Unclear situations with respect to property rights will be identified in the early stages before they create an obstacle for the project. Management of IPR will be taken seriously since the very beginning of the project.

Outcomes: IPR Management Plan.

Participation per Partner	
Partner number and short name	**WP1 effort**
1 - Ceit-IK4	
2 - LASEA	
3 - MULTITEL	
4 - VISUM	
5 - CICe	
CEGASA	
6 - FAGOR	

7 - RESCOLL	
8 - ESCI	
Partner number and short name	**WP1 effort**
Total	

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D1.1	Project Management Handbook	1 - Ceit-IK4	Report	Confidential, only for members of the consortium (including the Commission Services)	3
D1.2	Data Management Plan.	1 - Ceit-IK4	ORDP: Open Research Data Pilot	Confidential, only for members of the consortium (including the Commission Services)	6
D1.3	IPR Management Plan	1 - Ceit-IK4	Report	Confidential, only for members of the consortium (including the Commission Services)	6

Description of deliverables

D 1.1 Project Management Handbook. (M3)
D 1.2 Data Management Plan. (M6) D
1.3 IPR Management Plan. (M6)

D1.1 : Project Management Handbook [3]

This deliverable will summare all the information required for an optimum management of the project (in terms of administrative forms, procedures, templates, financial aspects, communication procedures, deliverable schedules, reporting periods, contacts, etc.), and conflicts resolution.

D1.2 : Data Management Plan. [6]

This deliverable will include details about the data management plan defined.

D1.3 : IPR Management Plan [6]

This deliverable will include information about the IPR Management Plan.

Schedule of relevant Milestones

Milestone number[18]	Milestone title	Lead beneficiary	Due Date (in months)	Means of verification

Work package number [9]	WP2	**Lead beneficiary [10]**	3 - MULTITEL

Work package title	DESIGN AND DEVELOPMENT OF A PROTOTYPE FOR CONTROLLED LIPSS		
Start month	1	End month	24

Objectives

The main objective of this WP is the development of an easy to integrate and control, compact and low cost optical module for the Laser Texturing Prototype. This objective is divided in the following specific objectives:

- To determine the detailed specifications of the optical module required for each application (material and functionalization required) to obtain the optimum performance for each one.
- To define the best beam shape(s) characteristics to cover a broad range of applications.
- To define the interfaces between the future optical modules and the Laser Texturing Prototype.
- To develop the optical module(s) to be integrated in WP5

Description of work and role of partners

WP2 - DESIGN AND DEVELOPMENT OF A PROTOTYPE FOR CONTROLLED LIPSS [Months: 1-24]
MULTITEL, Ceit-IK4, LASEA

In this WP the consortium (leaded by MULTITEL) will work to develop the optical module of the Laser Texturing Prototype. This optical module is the responsible to control the parameters that determine the LIPSS formation: optical power, beam profile, polarisation orientation, and even the operation wavelength of the micromachining system.

Task 2.1 Definition of the System Architecture.
Lead Partner: MULT - Participant Partners: LASEA, CEIT, VISUM, FAGOR, RESCOLL and CICe - Task duration: M01-M06

The partners involved in this task will work on determining the detailed specifications required for each material/ functionalization combination defined for the development of the Prototype, in order to obtain the optimum performance in terms of LIPSS formation quality and process time for each one. The more relevant specifications include: (i) Determination of the minimum and maximum line width, that is the width covered with LIPSS with a specific orientation. (ii) Determination of the range of energy densities necessary to cover the materials and functionalization combinations. (iii) Determination of the beam profile/shape. (iv) Determination of the polarization requirements (linear, circular, elliptical,…) (iv) Determination of the laser wavelength.

Taking into account the specifications defined, MULTITEL and LASEA will make a preliminary definition of the Laser and optic architecture needed to adequately cover the range defined.

Task 2.2 Preparation of the experimental laboratory set-up and tests.
Lead Partner: MULTITEL - Participant Partners: CEIT, LASEA - Task duration: M01-M15

In this task, an experimental laboratory set-up will be defined with the aim of testing and defining more precisely the parameters of the future optical module(s) defined in Task 2.1.

The goal will be to define the best beam shape(s) characteristics to cover a broad range of applications (materials and functionalities). Top-hat beam shape will be certainly the mostly used but depending on the applications requirements like working distance, spot size and depth of field, the exact characteristics of the top-hat profile may be adapted. In particular, the steepness and the homogeneity of the beam shape are of great importance. To define the best beam shape(s), MULTITEL will upgrade its current system.

The system that MULTITEL currently uses (see figure 13) is based on a milliJoule femtosecond IR source coupled to high speed galvanometric (scanning module). It comprises a spatial beam shaping module that includes a beam shaping based on a Diffractive Optical Element (DOE) and a polarization part, which permits to manually orientate the linear polarisation state of the beam at the input of the beam shaper (spatial beam shaping module).
See FIGURE 13 CURRENT SET-UP FROM MULTITEL TO BE ADAPTED FOR THE PROJECT in Appendix I of part B.

In order to set the required parameters, it is necessary to address a wide range of optical configurations. Therefore, flexibility is the most important parameter regarding the beam shaping method to be used, regardless of integration losses and handling power. In this regard, to carry out the experimental tests the currently system used by MULTITEL will be upgraded to obtain a more flexible system by:

- The implementation of a spatial light modulator (SLM) based polarization and beam shaper for producing controlledpolarized beams: linearly, circular, azimuthally and radially polarized beams. An optical arrangement including a SLM component will permit a more flexible control of the polarization, not limited to linear state. As for beam shaper, the use of use SLMs instead of DOEs will allow to change the beam shape, obtaining a flexible and dynamic beam shaping and control.
- The implementation of multi-beam splitting functions.

Based on this flexible and dynamic beam shaping and control system, the following experimental test will be carried out to define the exact parameters of the optical modules required for achieving the specifications defined in task 2.1: - First, for each material (Aluminium, cooper, titanium and stainless steels) the beam profile and polarization orientation to obtain different ripples will be defined (in terms of depth and periods) by: test of Gaussian, Super Gaussian and Tophat with different steepnesses combined with different polarization modes.
- Then, for each application (materials and functionalities) the best beam profile in terms of depth of focus, overlappingand sensitivity to stitching issues will be defined. For instance, to make hydrophobic surfaces, it is important to be very careful about stitching issues and in this case, a precise overlap control with well-defined homogeneous beam is required.

A series of at least 20 samples of 10* 10 cm² from each material will be used for these tests.

From the two series of experiments, a trade-off will be defined between optical module impact on process sensitivity. Once the exact beam shape parameters have been defined, then for the development of the optical module (task 2.4) a passive solution will be preferred for cost and power handling aspects.

Outcomes: Exact optical configuration of the optical module(s).

Task 2.3 Definition of the optical, electrical and mechanical interfaces of the optical modules
Lead Partner: LASEA - Participant Partners: MULT, CEIT - Task duration: M15-M18

The aim of this task is to define the interfaces between the future optical modules and the Laser Texturing Prototype. This includes software interface, electrical control, optical properties of the laser source in and out of the optical module and mechanical characteristics.

The different parameters to be defined are:
- Input beam characteristics: Beam size, Gaussian shape, M², spectral bandwidth, power and energy. Some requirementsdepend directly on the applications (like ablation thresholds will impose energy) but others are only dependant on the quality of the laser source. Although the optical module must be very robust to input laser parameters, some limits will be defined here. These limits will be settled from experimental tests at MULTITEL on the influence of the input beam shape on the output shape from the optical module.
- Mechanical dimensions of the optical module: This module will be integrated in the laser path for the galvanometricscanner. The weight, size and fixing/interfacing with the scanner will be defined here. A trade-off will have to be found between MULTITEL and LASEA regarding the compactness of the future system.
- Electrical interfacing: This corresponds mainly to the way that the polarization state will be driven and controlled. Theinput and output signals from/to the optical to/from the machine will be also defined between MULTITEL and LASEA.

The control panel will be integrated in LASEA global software.

Outcomes: Interfaces of the optical module.

Task 2.4 Implementation and test of the optical modules.
Lead Partner: MULT - Participant Partners: LASEA, CEIT - Task duration: M18-M24

In this task MULTITEL will construct a single universal optical module, able to address the whole panel of applications (materials/functionalities) defined for the project. The objective of the task is to obtain a low cost, compact solution, easy to integrate and control. In this case, dynamic flexibility will not be the main parameter for what concerns the beam shaping solution. As in most of industrial processes, dynamic beam shape is not required and induces extra-costs and can reduce the robustness of the system. If a generic beam profile can be found to cope with most of the applications, this will be the preferred industrial solution.

The aim of this task is to develop a low loss method, able to sustain high power and energy laser operation and robust to the input laser parameters fluctuation like in particular polarisation. Therefore, instead of Dynamic solutions, which can involve an important cost for beam shaping, Passive methods are going to be use in this solution. Among the different passive methods, diffractive optics will be probably used for beam shaping because of its compactness, robustness and easiness of integration. Nevertheless, if on the preliminary test performed in task 2.2 it is proven that a dynamic beam shaper is the most sensitive solution to cover the whole panel of applications, then at the begging of this task will take place a study about the economic impact of dynamic beam shape, analysing its cost against its potential market to determine its use.

The beam shaping element will be combined with a polarization rotating element that will be externally driven for a remote control of the beam polarization state. A schematic view of the system can be seen in Appendix I of part B. More than one module may be constructed depending on the applications requirements. A maximum of three versions will be prepared in order to cover the whole panel of applications defined. If during the definition phase it is identified that more than three versions are necessary, then the potentiality to implement a dynamically flexible beam shaping solution instead of optical modules based on passive beam shaping methods will be considered. In this case, the static solutions will be analysed during WP7 for the replication guidelines for specific applications where the requirements are less restrictive. This will be an important milestone of this WP at M18. The decision will depend on LASEA, MULTITEL and CEIT and will be based on solution costs, easiness of integration and robustness.

The optical module(s) developed will be evaluated in the laboratory experimental set-up prior integration in WP5. The evaluation will consist on texturing the same samples as in task 2.2 and comparison with previous results.

Outcomes: Optical module(s) to be integrated in WP5.

Participation per Partner

Partner number and short name	**WP2 effort**
1 - Ceit-IK4	
2 - LASEA	
3 - MULTITEL	
Total	

List of deliverables

Deliverable Number[14]	**Deliverable Title**	**Lead beneficiary**	**Type**[15]	**Dissemination level**[16]	**Due Date (in months)**[17]
D2.1	Report on detailed specifications from all the applications	2 - LASEA	Report	Confidential, only for members of the consortium (including the Commission Services)	6
D2.2	Report on optical configuration of the optical module(s)	3 - MULTITEL	Report	Confidential, only for members of the consortium (including the Commission Services)	15
D2.3	Report on definition of the interfaces of the optical module	2 - LASEA	Report	Confidential, only for members of the consortium (including the Commission Services)	18
D2.4	Optical modules prototypes to be integrated in WP5	3 - MULTITEL	Demonstrator	Confidential, only for members of the consortium (including the Commission Services)	24

Description of deliverables

D.2.1: Report on detailed specifications of the optical module (M06)
D.2.2: Report on optical configuration of the optical module(s) (M15)
D.2.3: Report on definition of the interfaces of the optical module (M18)
D.2.4: Optical modules prototypes to be integrated in WP5 (M24)

D2.1 : Report on detailed specifications from all the applications [6]
This deliverable will include details of the specifications of all the applications

D2.2 : Report on optical configuration of the optical module(s) [15]
This deliverable wil give details about the optical configuration of the optical module(s)

D2.3 : Report on definition of the interfaces of the optical module [18]
It will include information about the interfaces of the optical module

D2.4 : Optical modules prototypes to be integrated in WP5 [24]
Optical modules prototypes to be integrated in WP5

Schedule of relevant Milestones

Milestone number[18]	**Milestone title**	**Lead beneficiary**	**Due Date (in months)**	**Means of verification**
MS1	Identification of the type of beam shaping to be used (dynamic or passive)	3 - MULTITEL	18	Conclusions about analysis of costs, easiness of integration and robustness
MS2	Optical module	3 - MULTITEL	24	Validation of the module by laboratory experimental test

Work package number [9]	WP3	**Lead beneficiary** [10]	4 - VISUM
Work package title	DESIGN AND DEVELOPMENT OF AN IN-LINE MONITORING UNIT		
Start month	1	**End month**	24

Objectives

WP3 addresses the development of the in-line quality control system, following the idea that the industrial measurement technologies should progress in parallel to the manufacturing advancements.
The specific aims of the WP are:
- Developing the core optical technology suitable for nanometric surface features quality control.
- Developing the beta-version of the analysis software.
- Developing the functional prototype suitable for industrial settings use.
- Testing and validating the new nano-metrology prototype functionalities and performance.

Description of work and role of partners

WP3 - DESIGN AND DEVELOPMENT OF AN IN-LINE MONITORING UNIT [Months: 1-24]
VISUM, Ceit-IK4, LASEA, MULTITEL
Task 3.1 Core optical system development
Lead Partner: VISUM - Participant Partners: CEIT, LASEA - Task duration: M01-M07

Using surface textured samples that are available from previous research projects from CEIT and VISUM, VISUM will start very early in the project to work on the offline trials with a benchtop diffractometry test rig incorporating different illumination sources (Different wavelengths, visible and non-visible light) and polarisations, and compact optical designs with high numerical aperture collection optics, to validate the suitable set up for measuring samples, while manufacturing process analysis will be advanced in parallel. CEIT and LASEA will use their previously existing setups to fabricate and provide samples in order to cover the whole range of all the different material/functionality combinations. This means different LIPSS periods, depths and orientation as well as different materials.

Tests will be performed in the lab scale in non-continuous punctual mode on a relevant set of samples (between 20-30 samples will be analysed) for obtaining the textures' features of interest, on planar and curved surfaces. CEIT and VISUM will test the reference samples by electron microscopy and analysed by image analysis, including FFT (Fast Fourier Transform) to calibrate the optical systems. Where possible, simple and direct correlations will be made between structural features and diffracted light patterns, to accelerate the analysis process, for example relating diffracted order orientation with the direction nano-scale lines on the surface.

Outcomes: Identification of the optimum optical technology system to develop

Task 3.2 Data acquisition and processing software development
Lead Partner: VISUM (Third party IRIS) - Participant Partners: CEIT - Task duration: M06-M21

The software is intended for driving the instrument, processing the data by means of predictive models and delivering the results through the user interface and towards the production line operators. This task will be performed together with CEIT.

Hardware drivers and the raw data acquisition layer will be set up for the core device used in the previous task.

The right modelling package must extract the dimensional data from the diffractometry analysis performed in task 3.1. Optical modelling of the diffraction behaviour for TE (Transverse Electric) and TM (Transverse Magnetic) polarised light will be carried out using Rigorous Coupled Wave Analysis. A beta version including the required mathematical treatment of the spectra will be delivered for the lab test rig for each of the pattern types.

The final version, adapted for inline configuration, will build upon this initial version while incorporating all specific detailed analysis of the nano properties to analyse by a plant operator for being used as a user friendly inline monitoring tool to control the correct part quality. As such, a Graphical User Interface will be set up. The data from the monitoring software will feed in the models regarding the process influence on the textures. In order to establish the same communication channels, there will be an interaction with WP4 to assure the compatibility of the codes. This task will be done together with CEIT.
Upon post project integration (WP5), long term robustness trials, and further control system development, the communication capabilities for sending information to external systems (SCADAs, PLCs, etc.), for example, the production line rejection system, or process optimisation routines could be integrated in close loop.

Outcomes: User interface and software analysis for data acquisition

Task 3.3 Functional prototype development
Lead Partner: VISUM (Third party IRIS) - Participant Partners: LASEA - Task duration: M10-M21

Based on the results of the offline trials performed in task 3.1 but also on the final specifications of the upscaled LIPSS processes that will have been developed in parallel in WP5 on the LASEA premises, eg. in terms of part size, patterning speed, etc., the architecture of the system hardware will be conceptualized fulfilling the system specifications. The optical systems will be designed using ray-tracing software (e.g. ZEMAX). In addition, the enclosure and other optomechanical parts will be designed using mechanical CAD tools (e.g. SolidWorks). The components (detector, light, data acquisition hardware) will be selected among commercially available options whenever possible and be customised when necessary.

Flexible sample mounting and stabilisation platforms will be designed by IRIS to orient the samples in a suitable configuration to the diffractometry optics, including tailored inserts for the screw parts.

The possibility of developing a maximum of two different prototypes tailored to the different resolution needs will be considered, so that different in-line demonstrators will be obtained to perform functionality trials. This will prevent over-design and contain final measuring system costs for surface functionalisation of products where patterns are above 500 nm range, thereby enabling the use of inexpensive 405nm lasers which are a factor of 300 cheaper than UV lasers. The use of TE and TM polarised 405nm light is the preferred option to characterise structures smaller than 500nm, with a cost increase due to optical design complexity. Independently of the solution required for the LASER4SURF project, in WP7, the rest of alternatives will be included in the replication guidelines for less/more restrictive conditions.

Outcomes: Optical system prototype built based on initial specifications

Task 3.4 Nano-metrology lab testing and validation
Lead Partner: VISUM - Participant Partners: CEIT, LASEA - Task duration: M18-M24

Using the system from the previous task, and data processing algorithms developed based on task 3.1 off line trials, further tests will be done using a test rig for continuous/dynamic trials at VISUM This will allow acquiring new data for the final version of the software to account for the influence on the continuous measurement and of the part curvature, and to provide a sampling refresh rate required to provide sufficient coverage of the surface, sampling of small patterned areas, and the speed of the process. The models will be adjusted as needed to provide final calibration before its integration of the global prototype. Any other improvement required from the test will be performed both in terms of hardware and software. A user manual will be delivered together with the systems. Acceptance Test procedure will be implemented prior to the validation test work in WPs 5 and 6

Outcomes: In line quality control system built test and validated for the different user cases

Participation per Partner

Partner number and short name	**WP3 effort**
1 - Ceit-IK4	
2 - LASEA	
3 - MULTITEL	
4 - VISUM	
IRIS	
Total	

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D3.1	Design and delivery of the system for lab testing (hardware and software)	4 - VISUM	Demonstrator	Confidential, only for members of the consortium (including the Commission Services)	7
D3.2	User Manual of the monitoring system	4 - VISUM	Report	Confidential, only for members of the consortium (including the Commission Services)	24
D3.3	In-line monitoring module prototype to be integrated in WP5	4 - VISUM	Demonstrator	Confidential, only for members of the consortium (including the Commission Services)	24

Description of deliverables

D3.1 Design and delivery of the system for lab testing (hardware and software) (M07)
D3.2 User Manual of the monitoring system (M24)
D3.3 In-line monitoring module prototype to be integrated in WP5 (M24)

D3.1 : Design and delivery of the system for lab testing (hardware and software) [7]
System for lab testing (hardware and software)

D3.2 : User Manual of the monitoring system [24]
User Manual of the monitoring system

D3.3 : In-line monitoring module prototype to be integrated in WP5 [24]
In-line monitoring module prototype to be integrated in WP5

Schedule of relevant Milestones

Milestone number[18]	Milestone title	Lead beneficiary	Due Date (in months)	Means of verification
MS3	In-line monitoring module	4 - VISUM	24	Validation of the module by comparison with external characterization methods.

Work package number [9]	WP4	**Lead beneficiary** [10]	1 - Ceit-IK4
Work package title	MODELLING FOR TOOLS FOR LIPSS PROCESES AND PARAMETER SELECTION		
Start month	1	**End month**	24

Objectives

The aim of this WP is to develop a software tool that enables process parameters selection based on a database of LIPSS processes and their resultant functionalities. The specific objectives are the following:
- To generate a database of LIPSS processes and resulting functionalities as a function of laser system parameters

- To define an analytical expression of the LIPSS properties as a function of process parameters
- To design and develop a software platform that provides the process parameters as a function of the desired functionality- To validate the software application for process parameter selection

Description of work and role of partners

WP4 - MODELLING FOR TOOLS FOR LIPSS PROCESES AND PARAMETER SELECTION [Months: 1-24]

Ceit-IK4, LASEA, CICe, FAGOR, RESCOLL

The aim of this WP is to develop the necessary tools that enable industrial end-users to select the process parameters that give rise to a desired functionality, without a need to deeply understand the LIPSS generation process. This objective implies for main steps, as sketched in the next figure. Each of the tasks detailed below corresponds to each specific step in the modelling tool development.

It is important to highlight that, the modelling tool developed in this WP intends to be a general tool that can be upgraded upon finalization of the tool and/or project ending. This way, new product functionalities and materials can be added to the tool. To demonstrate this upgradable capability, WP7 will include additional LIPSS experiments for the new products/sectors identified for potential replication of the technology.

Task 4.1 Generation of a database of LIPSS processes and resulting functionalities

Lead Partner: CEIT - Participant Partners: LASEA, MULTITEL, CICe, FAGOR, RESCOLL - Task duration: M1-M24

The generation of LIPSS on metallic materials and their properties depend both on laser process parameters (wavelength, energy per pulse, pulse duration, number of accumulated pulses per area, polarization, scanning velocity …) and material properties (initial roughness, electric properties, optical properties …). As a result, LIPSS with different periods, depths and orientations can be obtained. Therefore, the first step of the software development is the generation of database that contains the obtained LIPSS geometries and their related functionalities, together with the process parameters used.

This will require the fabrication of LIPSS on the surface of the target materials (aluminium, copper, titanium and stainless steel) as well as a characterization of the resulting geometries and functionalities:

- First of all, an experimental matrix will be defined for each of the 4 selected materials and functionality pair. Eachmatrix will contain at least 7 values for each of the parameters affecting LIPSS formation (energy per pulse, number of pulses, scanning speed, beam polarization, etc.) These values will be selected according to end-user recommendations for the use cases (CICe, CEGASA, FAGOR, RESCOLL) and according to previous results and bibliographic search for the rest of functionalities. A series of 5 samples will be manufactured by CEIT for each combination of parameters. - The structural characterization of the resulting surfaces will be done by Scanning electron microscopy (SEM) and atomic Force microscopy in order to analyse the period, depth and orientation of the LIPSS by CEIT and CICe. A Fourier analysis of the SEM images will enable an accurate determination of the mean period of the structures and it will be performed by CEIT.
- The characterization of those samples specifically related to the project use cases will be evaluated according to eachspecific application (batteries, encoders and/or implants) by the partners involved in each of them according to their specific procedures (CICe, Fagor, RESC) in order to assess the resulting functionality. As the functionalization for batteries is performed in the current collectors and a number of fabrication steps are need afterwards, the evaluation of this particular case requires a deeper analysis. This particular evaluation will be performed as follows:

The characterization for the Batteries application field will include surface tension and wettability studies (against electrode slurry) to be carried out by CEIT and CICe in order to evaluate the coating ability of the slurries on the current collector surface. Untreated current collectors will also be characterized in order to evaluate the improvements obtained by LIPSS processing against untreated current collectors. CEGASA will define the best suited electrochemical testing

conditions required for the selected electrode materials to proof the concept and will also provide a detailed initial anode and cathode formulation and negative to positive mass/capacity ratio. Using this formulation as the starting point, CICe will work on the optimisation of electrode formulation with the help of CEGASA. The slurry preparation, electrode casting and calendering will be performed at CICe Dry Room facility considering the input provided by CEGASA whilst the electrochemical characterisation will require the participation of CICe and CEGASA. Discs will be punched from the different laminates for their electrochemical characterization which will be performed at CICe as half cells at a coin cell level. Once the half-cells are characterised the fabrication of full cells in coin cell format will start (CICe with advice from CEGASA); reference cells with non-structured and also with treated current collectors will be prepared. The electrochemical characterization for both half-cells and full cells will include C-rate capability tests, Cycle life tests and some tests at high temperature and different C-rates will be complemented by CEGASA when needed. Besides, structural, compositional and surface and interface analysis of pristine and cycled electrodes will be carried out by means of: SEM, XRD, Raman, FTIR and XPS. So as to understand the electrode evolution during electrochemical cycling and establish eventual correlations.
- The characterization for those samples related to other general functionalities will be evaluated at CEIT, CICe and FAGOR facilities. This includes the characterization of:
(i) Water contact angle for hydrophobic and anticondensation properties (CEIT)
(ii) Optical properties such as reflectance, absorbance, diffraction (FAGOR) (iii) Adhesion increase/decrease (CEIT, CICe)

Outcomes: Database of LIPSS processes and resulting functionalities as a function of laser system parameters

Task 4.2 Statistical analysis of the influence of process parameters on the obtained LIPSS and functionalities
Lead Partner: CEIT - Participant Partners: - Task duration: M6-M24

The information contained in the database generated in task 4.1 will be statistically analysed by CEIT in order to determine a) an analytical expression of the target properties of LIPSS (period, depth, orientation …) as a function of the process parameters; and b) the parameters that are more/less relevant to the structuring process and therefore to the resultant functionality.

Outcomes: Analytical expression of the LIPSS properties as a function of process parameters. Set of relevant and nonrelevant process parameters.

Task 4.3 Design and development of a software platform for parameter selection
Lead Partner: CEIT - Participant Partners: LASEA, MULTITEL- Task duration: M10-M18

Based on the results obtained in Task 4.2, a software platform will be designed and developed in order to enable parameter selection to industrial end-users. This SW application will ask the users for a desired functionality in a specific material and will provide the set of parameters necessaries for a successful process according to the study previously performed. The required parameters will be graphically presented to the user and will also be storage in a file that can be afterwards transmitted to the prototype tool in order to automatically program it.

Although other possibilities will be explored during the project, the mathematical capabilities and graphic user interfaces of Matlab© make this software a suitable candidate for the development of the tool.

The SW development will be mainly carried out by CEIT, although close collaboration with MULTITEL and LASEA will be necessary in order to establish the interfaces between subsystems and the way the information must be storage and transferred to the main control SW (the SW that controls the Optics module and the whole prototype)

Outcomes: A software platform that provides the process parameters as a function of the desired functionality.

Task 4.4 Pre-validation of the software tool
Lead Partner: CEIT - Participant Partners: LASEA, MULTITEL - Task duration: M19-M22

The software tool developed in task 4.3 will be pre-validated before its integration in the whole prototype. A set of target functionalities will be selected and introduced in the developed application. The process parameters determined by the application will then be configured in a laser processing equipment and the resultant LIPSS will be compared with the target ones. This pre-validation will be useful to detect possible mismatches before the integration in the final prototype and therefore minimizing risks in it.

Outcomes: Optimized and validated software application for process parameter selection.

Participation per Partner

Partner number and short name	WP4 effort
1 - Ceit-IK4	

2 - LASEA	
5 - CICe	
CEGASA	
6 - FAGOR	
7 - RESCOLL	
Total	

List of deliverables

Deliverable Number[14]	**Deliverable Title**	**Lead beneficiary**	**Type**[15]	**Dissemination level**[16]	**Due Date (in months)**[17]
D4.1	Database of LIPSS processes and related functionalities	1 - Ceit-IK4	Other	Confidential, only for members of the consortium (including the Commission Services)	24
D4.2	Report on process parameter determination	1 - Ceit-IK4	Report	Confidential, only for members of the consortium (including the Commission Services)	24
D4.3	Preliminary Software platform for parameter selection	1 - Ceit-IK4	Other	Confidential, only for members of the consortium (including the Commission Services)	22
D4.4	Optimized and validated software platform	1 - Ceit-IK4	Other	Confidential, only for members of the consortium (including the Commission Services)	24

Description of deliverables

D.4.1 Database of LIPSS processes and related functionalities (M24)
D.4.2 Report on process parameter determination (M24)
D.4.3 Preliminary Software platform for parameter selection (M22) D.4.4 Optimized and validated software platform (M24)

D4.1 : Database of LIPSS processes and related functionalities [24]

Database of LIPSS processes and related functionalities

D4.2 : Report on process parameter determination [24]

This deliverable will include details about process parameter determination

D4.3 : Preliminary Software platform for parameter selection [22]

Software platform that provides the process parameters as a function of the desired functionality.

D4.4 : Optimized and validated software platform [24]

Schedule of relevant Milestones

Milestone number[18]	Milestone title	Lead beneficiary	Due Date (in months)	Means of verification
MS4	Optimized and Validated Software platform for parameter selection	1 - Ceit-IK4	22	Functional validation prior to integration

Work package number [9]	WP5	**Lead beneficiary** [10]	2 - LASEA
Work package title	INTEGRATION OF THE DEVELOPED TOOLS		
Start month	1	**End month**	33

Objectives

The aim of this task is to develop a system that will merge in one single machine the optical module, inspection unit and modelling tool. The specific objectives are the following:
- To define the specifications at mechanical, optical and electric level of the integrated prototype.
- To ingrate the three modules developed in the previous WPs into a global system to validate the new concept ofmachine proposed in the project.
- To test the system and validate it.

Description of work and role of partners

WP5 - INTEGRATION OF THE DEVELOPED TOOLS [Months: 1-33]
LASEA, Ceit-IK4, MULTITEL, VISUM, CICe, FAGOR, RESCOLL
The integration of the different modules will be done using an already existing setup of LASEA. It consists of a granite based setup with a gantry. It can be equipped with a femtosecond laser emitting at 1030 nm with pulse durations of about 400 fs or with a nanosecond laser with wavelengths of 1064nm or 532 nm and pulse durations of about 100ns down to 15ns, bringing flexibility to the possibilities of the setup.

In this WP, the specifications for the integration of each module in the current setup will be defined for the different modules, taking into account the current technical aspects of the setup such as mechanical dimensions. This information will be gathered and communicated to other partners prior to the building of the modules in order to be taken into account in their designs.

Due to their roles and technical characteristics, the flexibility in the integration of the different modules will be different for each module. For instance, the module developed in WP2 is an inline module and will have the strictest limitations in terms of integration, directly related to the already existing configuration such as the dimensions of the granite, the gantry, which will bring some limitations on the possibilities of change on the mechanical parts such as the position of mirrors mounts and so, limitations on the change of the optical beam path. Regarding the monitoring module developed in WP3, those restrictions are less strict as this module is off beam path, and so there is more flexibility to integrate it on the setup, but still, in accordance with the configuration of the setup and the dimensions of the module, its location will be chosen depending on mechanics and optics aspects that will have to be defined accurately.

Those difficulties will be overcome thanks to a close collaboration between LASEA and the different partners involved in the WP2, WP3 and WP4.

Task 5.1 Study of the integration
Lead Partner: LASEA - Participant Partners: MULT, VISUM (and its Third party IRIS), CEIT, FAGOR, RESCOLL, Cegasa (TP of CICe) - Task duration: M01-M12

In this task LASEA will work on the design of the integration of the different modules to be developed in the WPs 2, 3 and 4 on the already existing setup.

For each module, an exchange of technical information between LASEA and those partners in charge of the development of the modules (MULT, VISUM and CEIT) will be done, in order to take into consideration the different interfaces between the modules and the station at mechanical, electrical, optical and software level. A document gathering all this information will be established. In this document, LASEA will determine the different technical aspects of the interfaces, which will be considered for the development of each module in WP2, WP3 and WP5 as well as for the development of the required changes in the existing setup:
- Mechanical: design and description of the components, their dimensions, their position, weight and way of fixing willbe described. Technical drawings will need to be exchanged between the partners in charge of the modules and LASEA. At the end, the complete design of the setup including the different modules will be defined.
- Optical: the characteristics of the beam will be provided such as the pulse energy, pulse duration, average power,positions and size of the waists, beam path.

- Electrical: this interface includes the power supply to be provided by the setup to the different modules, and the setup'selectrical interfaces. It also includes the communication protocols that it will be necessary to establish between the setup and the different modules. The way of control every functionality of the modules will be described both in terms of electronics and software requirements.
- Software: This interface includes the control of the different functionalities of every module integrated into the homemade software. It describes the way the different components are controlled such as the protocol of communication, commands to be sent, I/O, etc.

Industrial end-users' point of view will be also taken into consideration, as their recommendations and restrictions for their particular use cases will provide a worst case scenario for the LASER4SURF prototype. This will include aspects such as operational environment and operators perspectives among others.

Outcomes: Document defining the specifications of the setup, the optical module and monitoring modules. Technical documents of the integration: electrical drawings, mechanical design, software control for all modules.

Task 5.2 Integration of the different modules
Lead Partner: LASEA - Participant Partners: MULT, VISUM (and its Third party IRIS), CEIT - Task duration: M06-
M27

The goal of this task is to add the different modules to the existing setup following the design defined in task 5.1 determined by the different partners. As said in the task 5.1, some restrictions due to the use of an already existing setup might have been highlighted, leading to the necessity to design and modify some mechanical parts.

Subtask 5.2.1: Adaptation of the mechanical design, electrical and software
New elements of the existing setup will be designed according to the requirements defined in the T5.1 in order to adapt all the elements of the different modules. This will be done in close collaboration with the different partners in charge of the WP2, 3 and 4 and according to what will have been defined as necessary in those WPs. The software aspect will require a lot of development. Indeed, it will be necessary to control the different modules from WP2 and 3 and to integrate them into the home-made software Kyla, which is already included in all the setups and machines of LASEA. This LASEA made software combines the different controls for the different devices included on every setup and machine made in LASEA.

Subtask 5.2.2: Technical integration of the modules
In this subtask, once the adaptations of the current setup have been done, the different modules will be added. Some checks to the different elements will be performed to verify every functionality of each module at technical level. Those checks will allow to compare the specifications of each module (already integrated in the global setup) with the specifications defined in the WPs 2, 3 and 4. Due to the use of a different setup with a different configuration than the ones used to develop the modules in WP2, WP3 and WP4, some difficulties to repeat the specifications of every module are expected to be encountered at the first time. Several iterations with the task 5.1 might be necessary in order to adapt every module to the setup in terms of mechanics, optics, software, during which the technical changes will be guided by the difference of specifications obtained in WPs 2, 3 and 4 and in this task.

Outcomes: Integration of the three modules in an already existing working station (integrated prototype)

Task 5.3 Tests and validation
Lead Partner: LASEA - Participant Partners: MULT, VISUM (and its Third party IRIS), CEIT - Task duration: M18-
M33

In this task a series of tests will be performed and carried out with the goal of qualifying and establishing the reliability of the different parts of each module. This set of tests will reproduce the same tests as those carried out for the validation of each module:
- For the optical module, the same tests than the qualification tests of T2.2 will be reproduced and compared to theresults obtained in the same task. An analysis of this comparison will be done to point out any difference with the results obtained at MULTITEL in WP2.
- For the inline monitoring unit, the same tests than in tasks 3.1 and 3.4 will be performed after the integration of theunit on the setup. A comparison with the results obtained in WP3 will be done in order to assess on the performances of the module.
- The database created in WP4 will be validated by CEIT, testing the process parameters provided for certain predefinedsets of substrates and process results. The suitability with the laser system control software of Lasea (Kyla®), in order to be able to achieve an automated supply of process parameters into the machine jobs, will be verified by LASEA.

Once the integration of the different modules has been validated, the prototype will be used to carry out the functionalisation of the products of each use cases foreseen in WP6.

Based on the results of the validation tests that will be done in WP6 to analyse the functionalisation obtained, LASEA, MULT, VISUM and CEIT in collaboration with the end-users will evaluate in this task if any technical adjustment is required on the global prototype.

Outcomes: Validation of the integrated prototype. The independent and combined control of the different modules on the integrated prototype validated.

Participation per Partner	
Partner number and short name	**WP5 effort**
1 - Ceit-IK4	
2 - LASEA	
3 - MULTITEL	
4 - VISUM	
IRIS	
5 - CICe	
CEGASA	
6 - FAGOR	
7 - RESCOLL	
Total	

List of deliverables

Deliverable Number[14]	**Deliverable Title**	**Lead beneficiary**	**Type**[15]	**Dissemination level**[16]	**Due Date (in months)**[17]
D5.1	Definition of setup specifications of the modules in the global prototype	2 - LASEA	Report	Confidential, only for members of the consortium (including the Commission Services)	12
D5.2	Integration of the three modules in an already existing working station	2 - LASEA	Demonstrator	Confidential, only for members of the consortium (including the Commission Services)	24
D5.3	Report on Tests and validation	2 - LASEA	Report	Confidential, only for members of the consortium (including the Commission Services)	33

D5.4	Public report Conclusions on LASER4SURF system validation	2 - LASEA	Report	Public	33

Description of deliverables

D5.1: Definition of setup specifications of the modules in the global prototype (M12)
D5.2: Integration of the three modules in an already existing working station (M24)
D5.3: Report on Tests and validation (M33)
D5.4: Public report - Conclusions on LASER4SURF system validation (M33)

D5.1 : Definition of setup specifications of the modules in the global prototype [12]

This deliverable will include information about the setup specifications of the modules in the global prototype

D5.2 : Integration of the three modules in an already existing working station [24]

Integration of the three modules in an already existing working station

D5.3 : Report on Tests and validation [33]

This deliverable will explain the tests performed to qualify and establish the reliability of the different parts of each module and to validate the integrated prototipe.

D5.4 : Public report - Conclusions on LASER4SURF system validation [33] Public report with the conclusions on LASER4SURF system validation.

Schedule of relevant Milestones

Milestone number[18]	**Milestone title**	**Lead beneficiary**	**Due Date (in months)**	**Means of verification**
MS5	Operational validation of the global prototype	2 - LASEA	27	Functional test validation prior to carried out use case functionalization.

Work package number [9]	WP6	**Lead beneficiary** [10]	6 - FAGOR
Work package title	TECHNOLOGY VALIDATION IN THREE FIELDS		
Start month	18	**End month**	36

Objectives

The aim of this WP is to demonstrate the feasibility of the proposed laser texturing prototype in a set of 3 products of different sectors: Machine tool industry, healthcare sector and batteries. The specific objectives are the following:
- To obtain a current collector with enhanced adhesion and roughening using the laser texturing prototype -
To obtain a dental implant with adhesion and osteointegration properties using the laser texturing prototype
- To obtain an optical encoder with enhanced optical properties of reflection.

Description of work and role of partners

WP6 - TECHNOLOGY VALIDATION IN THREE FIELDS [Months: 18-36]
FAGOR, Ceit-IK4, LASEA, CICe, RESCOLL
Each of the task detailed below includes the specific activities that will be necessary to carry out to validate the product surface functionalisation on each use case.

It is important to highlight that the samples will be manufactured in LASEA premises and will be sent to the corresponding use cases partner who will carried out the necessary test to validate the functionalisation obtained. At least two sets of samples will be fabricated for each use case, although additional sets can be manufactured if necessary for the functionality validation tasks. The results of the different tests that will be performed in each task will be transfer to WP5, in order to analyse them and take into consideration any possible changes that would be required to guarantee the feasibility of the prototype.

Task 6.1 Technology Validation in Energy Storage.
Lead Partner: CICe - Participant Partners: CEGASA (third party), LASEA - Task duration: M18-M36

Subtask 6.1.1 Fabrication of structured current collectors for pouch cell prototypes
Lead Partner: CICe - Participant Partners: LASEA, CEGASA - Task duration: M18-M33
In this task LASEA will fabricate structured current collectors from commercial rolls used in battery industry following input from CICe and CEGASA. The total structured surface will allow for the fabrication of a battery prototype in pouch cell format.

Subtask 6.1.2 Monolayer pouch cell fabrication and electrochemical characterization.
Lead Partner: CICe - Participant Partners: CEGASA (third party) - Task duration: M22-M33
CICe will prepare monolayer pouch cells using electrodes which will be prepared using the structured current collector and also non-structured as reference. For this several parts of CICe's prototyping line will have to be adjusted. Coated area will be increased by a factor of ca. 20, which will represent a significant increase in the capacity delivered by the cell. Pouch cells are one of the different cell designs which are commercially available together with the cylindrical and prismatic configurations. Building pouch cells involves processing steps different from those to build coin cells such as rectangular electrode notching, ultrasonic welding, pouch sealing, degassing… which are common processing steps for commercial battery fabrication. Therefore, building a monolayer pouch cell will constitute a step ahead towards the fabrication of cells with standardized configuration and processing techniques to which it will be a closer comparative in terms of capacity and processing.

Subtask 6.1.3 Characterisation of pouch cell electrodes
Lead Partner: CICe - Participant Partners: CEGASA (third party) - Task duration: M31-M36
The electrodes to be used for the pouch cells will be initially characterised to ensure they meet the required standard and also to check reproducibility from coin cell up to pouch cell level. In addition, cycled pouch cells will be disassembled after use so as to study the electrode evolution. This characterisation will include visual inspection of the electrode and also some advanced characterisation such as SEM and XPS.

Outcomes: Current collector with enhanced adhesion and roughening, leading to:
- Improvement of a 30% in the life cycle of battery electrodes, compared to electrodes casted on conventional currentcollectors.
- Increase the battery C-rate capability by 20% when cycling at 1C and by 60% when cycling at 2C.

Task 6.2 Technology Validation in Biomedical Implants

Lead Partner: RESCOLL - Participant Partners: LASEA - Task duration: M18-M36

Subtask 6.2.1 Fabrication and dimensional characterization of modified implants
Lead Partner: LASEA - Participant Partners: RESCOLL - Task duration: M18-M33
The developed prototype will be used to modify the surface finishing of an implant fabricated with standard techniques (RESCOLL). The dimensional characterization of the surface performed by the in-line monitoring unit will be compared to the one performed by optical and scanning electron microscopes. The implants to be treated will be produced by RESCOLL Manufacturing in titanium allow (Ti6Al4V) using standard milling and cutting methods. RESCOLL Manufacturing will validate the samples prior to treatment using its standard QA/QC protocols (visual inspection using a KEYENCE IM-6225, non-contact dimensional measurement using a MMT VERTEX).

Subtask 6.2.2 Mechanical and chemical analysis and validation of the modified implants
Lead Partner: RESCOLL Participant Partners: -; - Task duration: M28-M36
The modified implants resulting from task 6.2.1 will be analysed and validated with the same techniques used for nonmodified implants in order to ensure that the effect of surface modification on the mechanical and chemical properties of the implants remain within the required ranges. Surface characteristics will be validated using profilometry and MEB, and compared to baseline implants prepared using traditional methods.
The validation of the implants insofar as mechanical and chemical properties will be performed in accordance with industry standards. The screws will be validated in accordance with ISO 18130, ISO 13498, and ISO 14801.

Subtask 6.2.3 Biological validation of the added functionality
Lead Partner: RESCOLL - Task duration: M30-M36
Subcontracted to Phycher, a risk assessment of the modified implants as compared to that of the standard implants will be undertaken. Particular attention will be made to biofilm formation, and whether or not there is an effect upon the rate of biofilm formation by the laser texturization. In addition, osteoblast/osteocyte compatibility will be validated in order to elucidate the effect (if any) of the laser texturization upon osteointegration.

Outcomes: Cellular adhesion and osteointegration of implants.

Task 6.3 Technology Validation in Linear Encoders
Lead Partner: FAGOR - Participant Partners: CEIT, LASEA - Task duration: M19-M36

Subtask 6.3.1 Design and Implementation of a validation setup
Lead Partner: FAGOR - Participant Partners: - Task duration: M19-M27
A testing setup actually located at Fagor's facilities and dedicated to the development of new products will be adapted in order to validate the application of LIPSS to linear encoders' fabrication. It will be necessary to add a reading terminal in a reflection configuration, instead of the actual one, based on transmission.

Subtask 6.3.2 Fabrication and dimensional/optical characterization of diffraction gratings
Lead Partner: LASEA - Participant Partners: FAGOR, CEIT - Task duration: M28-M36
The laser prototype resulting from WP5 will be used for the fabrication of diffraction gratings in order to validate it in the metrology application field. The fabricated gratings will be dimensionally and optically characterized, both with the developed in-line monitoring (LASEA) unit and with external equipment: optical microscopy will be done by FAGOR and SEM/AFM, done by CEIT.

Subtask 6.3.3 Validation of the fabricated gratings in optical encoders
Lead Partner: FAGOR - Participant Partners: CEIT - Task duration: M28-M36
The functionality of the fabricated gratings will be tested in the validation setup developed in task 6.3.1. FAGOR will evaluated the more relevant parameters for each grating: obtained signal intensity, contrast and an estimation of the precision achieved with the grating. CEIT will correlate the gratin properties with the surface LIPSS.

Outcomes: Optical encoders with enhanced optical properties for reflection, leading to:
- To increase twofold the tolerances of distances between the grating and the optical head
- To eliminate the background light of the gratings

Participation per Partner

Partner number and short name	**WP6 effort**
1 - Ceit-IK4	
2 - LASEA	
Partner number and short name	**WP6 effort**

5 - CICe	
CEGASA	
6 - FAGOR	
7 - RESCOLL	
RMG	
Total	

List of deliverables

Deliverable Number[14]	**Deliverable Title**	**Lead beneficiary**	**Type**[15]	**Dissemination level**[16]	**Due Date (in months)**[17]
D6.1	Current collector functionalized and validated	5 - CICe	Demonstrator	Confidential, only for members of the consortium (including the Commission Services)	36
D6.2	Dental implant functionalized and validated	7 - RESCOLL	Demonstrator	Confidential, only for members of the consortium (including the Commission Services)	36
D6.3	Optical encoder functionalized and validated	6 - FAGOR	Demonstrator	Confidential, only for members of the consortium (including the Commission Services)	36
D6.4	Summary report on test performed on batteries	5 - CICe	Report	Confidential, only for members of the consortium (including the Commission Services)	36
D6.5	Summary report on test performed on implants	7 - RESCOLL	Report	Confidential, only for members of the consortium (including the Commission Services)	36
D6.6	Summary report on test performed on encoders	6 - FAGOR	Report	Confidential, only for members of the consortium (including the Commission Services)	36

Description of deliverables

D6.1: Current collector functionalized and validated (M36)
D6.2: Dental implant functionalized and validated (M36)
D6.3: Optical encoder functionalized and validated (M36)
D6.4: Summary report on test performed on batteries (M36)
D6.5: Summary report on test performed on implants (M36)
D6.6: Summary report on test performed on encoders (M36)

D6.1 : Current collector functionalized and validated [36]
Current collector functionalized and validated

D6.2 : Dental implant functionalized and validated [36]
Dental implant functionalized and validated

D6.3 : Optical encoder functionalized and validated [36]
Optical encoder functionalized and validated

D6.4 : Summary report on test performed on batteries [36] This deliverable will detail the test performed on batteries.

D6.5 : Summary report on test performed on implants [36]
This report will detail the test performed on implants

D6.6 : Summary report on test performed on encoders [36]
This deliverable will detail the test performed on encoders

Schedule of relevant Milestones

Milestone number[18]	Milestone title	Lead beneficiary	Due Date (in months)	Means of verification
MS6	Feasible surface functionalization of use cases	6 - FAGOR	36	Tests completed and approved related to each use case.

Work package number [9]	WP7	**Lead beneficiary [10]**	1 - Ceit-IK4
Work package title	GATHERING EXPERIENCE AND NEW SCENARIO DEFINITION		
Start month	25	**End month**	36

Objectives

With the aim of maximising the future impact of the project results, this WP aims to compare the results obtained with the expected results in order to define new strategies for an effective replication of the process developed in other areas of application within the sectors covered by the use-cases as well as in other sectors. The specific objectives are the following:
- Create replication guides
- Identify potential replication sectors
- Maximize the impact of the project results

Description of work and role of partners

WP7 - GATHERING EXPERIENCE AND NEW SCENARIO DEFINITION [Months: 25-36]
Ceit-IK4, LASEA, MULTITEL, VISUM, CICe, FAGOR, ESCI
Task 7.1 Use case analysis
Lead Partner: CEIT - Participant Partners: CICe, LASEA, MULTITEL, CEGASA (TP of CICe), RESCOLL, FAGOR - Task duration:M25-M33

In this task, we will benchmark the results from WP5 and WP6 against state-of-the-art technology. The benchmark will include technical and economic aspects of the targeted cases:
- Technical parameters:

(i) Batteries: C-rate and increase of life cycles (analysis performed by CICe and CEGASA)
(ii) Implants: polarizability of cells and increase of cell density (analysis performed by RESCOLL)
(iii) Metrology: Increase in signal contrast and increase in positioning tolerances between the gratings of the system(analysis performed by CEIT and FAGOR).
- Economic aspects: three parameters will be calculated for the techno-economic evaluation:
(i) The total cost of the addition of the new process will be calculated for the 3 cases analysis in WP6 (analysis performedby LASEA and CEIT).
(ii) The marginal increase in performance per euro will be calculated (analysis performed by RESCOLL, FAGOR, CEGASA)
(iii) The required minimum performance for commercialization will be evaluated (analysis performed by RESCOLL, FAGOR, CEGASA)

The results of these analysis will be discussed in terms of technical maturity (TRL), degree of achievement of the objectives and further economic requirements (if any).

Task 7.2 Identification of potential replication sectors
Lead Partner: LASEA - Participant Partners: CEIT, CEGASA, RESCOLL, FAGOR - Task duration: M28-M32

Two different approaches for potential replication and future applications will be identified here:
- Application of LASER4SURF technology to other sectors than the targeted ones (LASEA, CEIT).
The replication of the technology for building, transport and manufacturing sectors will be analysed with special attention. The different applications and potential replication sectors will be identified according to the following parameters: (i) Accessibility; (ii) Additional functionality; (iii) Added value of the process; (iv) Economic impact. - Further applications in the three targeted areas (Energy, Biomedical, Metrology) will be considered (CEGASA, RESCOLL, FAGOR).
Based on the results on WP6, and taking into account the technical aspects of the prototype and the processes that have been developed and implemented, the end-users will evaluate the replication potential of LASER4SURF technology in different products and/or applications of their expertise fields.

Task 7.3 Replication guides
Lead Partner: CEIT - Participant Partners: MULTITEL, VISUM, LASEA - Task duration: M32-M36

In this task a technical analysis will be performed and a recommended guide will be developed to further target the most interesting sectors identified in task 7.2. The analysis will include the target materials, the typical dimensions and shape of the pieces, fabrication tolerances and requirements. Although the functionalization prototype developed in the first phase of the project will be a general prototype covering a wide range of functionalities and specifications, certain modifications of the developed modules (optical module and control, monitoring unit and parameters selection tool) may be required for the new target products and sectors, and therefore a guideline for these modifications will also be defined. The necessary modifications may include:
- Optics Module: Different laser wavelengths, repetition rates and/or energies may be required for certain applicationsand/or materials in order to reach specific processing times. Therefore, optical components like lenses and mirrors should be replaced. This study will be performed by MULTITEL
- Monitoring Unit: A replacement of the light source may be necessary in order to adapt the module to different structuressizes. This will also require an adaptation in the processing software. This study will be performed by VISUM
- Parameter Selection Tool: the developed tool will be an upgradable solution and therefore will enable the modificationof the database in order to provide a solution for materials and functionalities different from the ones studied in WP4. In this context, additional experimental work will be performed (following the experimental description of tasks 4.1 and 4.2) in order to extend the database to metallic alloys and functionalities determined in task 7.2. This study will be performed by CEIT and CICe.

Participation per Partner	
Partner number and short name	**WP7 effort**
1 - Ceit-IK4	
2 - LASEA	
3 - MULTITEL	

4 - VISUM	
5 - CICe	
6 - FAGOR	
8 - ESCI	
Total	

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D7.1	Benchmark tables and analysis of the use cases	1 - Ceit-IK4	Report	Public	33
D7.2	List of potential replications sectors and new applications	1 - Ceit-IK4	Report	Public	36

Description of deliverables

D7.1: Benchmark tables and analysis of the use cases (M33)
D7.2: List of potential replications sectors and new applications. Description of the applications and discussion of the market and technical opportunity, including the guidelines for replication (M36)

D7.1 : Benchmark tables and analysis of the use cases [33]

Public deliverable with benchmark tables and analysis of the use cases

D7.2 : List of potential replications sectors and new applications [36]

Public deliverable with the list of potential replications sectors and new applications

Schedule of relevant Milestones

Milestone number[18]	Milestone title	Lead beneficiary	Due Date (in months)	Means of verification
MS7	Potential replications sectors and new applications identified	1 - Ceit-IK4	36	Replication guides agreed and prepared

Work package number [9]	WP8	**Lead beneficiary** [10]	2 - LASEA
Work package title	DISSEMINATION AND EXPLOITATION		
Start month	1	**End month**	36

Objectives

This WP pursues two parallel objectives: preparing the exploitation of the project knowledge and informing about the project and its achievements. To this end, this WP will:
- Detail the future exploitable results of the project and develop exploitation strategies for each of them...
- Guarantee an optimum knowledge management including IPR aspects.
- Disseminate project results to relevant stakeholders in industry, academia and policy making circles.

Description of work and role of partners

WP8 - DISSEMINATION AND EXPLOITATION [Months: 1-36]
LASEA, Ceit-IK4, MULTITEL, VISUM, CICe, FAGOR, RESCOLL, ESCI
Task 8.1. Exploitation
Lead Partner: LASEA - Participant Partners: CEIT, MULT, VISUM, FAGOR, RESCOLL, CIC - Task duration: M7-M36

The task will implement a specific methodology provided by the partner (LASEA) based on a series of specific tasks that will enable the optimization of the exploitation possibilities of the project results.

Subtask 8.1.1 Identification of key results and functional analysis of each key result. (LASEA, CEIT, MULTITEL, VISUM, FAGOR, RESCOLL, CIC)
This task will carry out a series of workshops aimed at identifying the most promising results for exploitation. The project outputs with the greatest potential for exploitation are already identified at this proposal stage, but the workshops will make the revision of these expected outputs adjusting them (if necessary) to the reality of the R&D being performed, and already keeping in mind the exploitation objectives of the running project.

After the identification of the key results, the task will carry out a functional analysis of each of the identified results. This functional analysis will identify what is the characteristic or ability of the different outputs to meet the end users' needs according to their feedback (the presence in the consortium of industrial partners such as RESCOLL, FAGOR and CEGASA (Third party of CICe) will help). The reason for making this analysis is to ensure that the project delivers results that match the purpose for which they have been designed, adequately answering the needs of the end users. The task will identify and characterise the functions that the project outputs should offer to meet the needs of the users.

Subtask 8.1.2: Value innovation analysis and market analysis. (LASEA)
After the functional analysis of the key results, a value innovation analysis will be carried out, which means identifying the features, benefits and advantages of the key outputs. The objective of the project is to provide a product that maximizes these advantages to be compelling to the end user. Moreover, the characteristics of the product should make it competitively distinctive, so that the commercial exploitation possibilities are maximized. According to this, the value innovation analysis will try to set the stage for answering the next question "How to make the project outputs significantly unique or competitively advantageous?"

Within the task, the real value of the expected outputs for the users will be examined in detail. It is necessary to compare the value of the project outputs with the potential market competitors, and this is why a market analysis will be carried out. In the framework of this market analysis, the following subtasks will be considered: Analysis of the potential customer demands, Identification and characterisation of the potential customers, why and how they would buy the product or service?; Market size, trends and opportunities; Analyse competing solutions; What are the competing solutions available on the market today?; How well do they match the functional requirements that we have identified?; What is their customer price?

Subtask 8.1.3: Value proposition and value proposition testing. (LASEA, CEIT, MULT, VISUM, FAGOR, RESCOLL, CICe)

Once the value innovation analysis has been completed, the next task will be to make a new innovation value proposition for the expected project outputs, which optimizes the value of the project outputs in comparison with potential competitors. This means building and proposing a "compelling" and competitively distinctive value curve so that we make sure that the project outputs stand out of the competition while still being outputs achievable within the project scope.

The value proposition will be built with the objective of making the project outputs significantly unique or competitively advantageous. For this, four directions will be explored for making the value innovation proposal:

- To cancel value: Which of the value components that competitors take for granted should be eliminated?
- To reduce value: Which value components can be reduced below the actual standards of industry?
- To increase value: Which value components must be increased above the actual standards or industry?- Which value components, that competitors have never offered, should be created?

Within this task, it will be decided which combination of the described directions will be taken in order to make the project value innovation proposition.

Once the value innovation proposition has been established, this will be tested with potential end users or early adopters to assure that it will answer their needs and expectations. This will be accomplished through interviews and questionnaires to a set of identified potential end users. The result of this testing will be fed-back to the value proposition in order to tune it according to the end users' requirements.

Subtask 8.1.4: IPR analysis of the new knowledge and project outputs. (LASEA, CEIT, MULT, VISUM, FAGOR, RESCOLL, CIC)
This task will deal with the IPR issues related to the project exploitation, and includes: (i) The definition of pre-existing knowledge of the project partners (Background), including a mapping of existing patents and potentially overlapping IPR; (ii) An assessment of the knowledge generated in the project (Foreground). (ii) A proposition of optimal IPR protection options, in line with the selected business model options (to be selected in the next task); (iii) A mapping of existing relevant standards and of standards in development; (iv) Considerations with regards to the standardisation strategy (e.g. get involved in on-going standardisation technical committees or develop a new standard…)

Subtask 8.1.5: Development of business models, business plans and exploitation action plans. (LASEA, CEIT, MULT, VISUM, FAGOR, RESCOLL, CIC)
Within this task the different business options for the project results will be investigated, establishing the preliminary business models. These potential business model options may include: Take-up by one of the partners (or a joint-venture of partners) and commercialisation of the output by this partner with adequate distribution channels; Patenting of the output with third-party licensing; Direct commercialisation of the output; Integration of the output in another solution. According to the selected business models, the task will build the exploitation action plans for each of the results of the project. The structure of the exploitation plans will include: The consortium and the opportunity; The solution and value proposition; Market and competition analysis; Business model options and marketing plan (distribution architecture / sales channels, customer relationship); Resources for implementing exploitation (key activities and value chain, key capabilities, partners); Financial planning (cost and revenues); Timeline; Risk management and contingencies; IPR strategy + Standardisation strategy.

Task 8.2. Communication & Dissemination
Lead Partner: ESCI - Participant Partners: All partners - Task duration: M1-M36

This task includes the following subtasks:
Subtask 8.2.1. Communication Master Plan (LASEA, CEIT, MULT, VISUM, FAGOR, RESCOLL, CIC)
The 'Communication Master Plan' outlines the communication and dissemination activities planned by the LASER4SURF partners for the whole project duration. Produced in M3, this document describes the overall communication strategy for the consortium, and functions as a guide for project partners when speaking about or on behalf of the project. This baseline document for communication and dissemination activities will be updated once during the project lifetime in M20. Additional information about the issues included in the "Communications Master Plan" is provided in section 2.2.1.

Outcomes: Communication Master Plan defined and distributed among partners.

Subtask 8.2.2. Visual Identity (ESCI)
A logo and payoff will be developed in the very initial phase of the project to create an identity that will be specific for LASER4SURF actions and partners, and to distinguish the project from other projects. Outcomes: Visual identity defined.

Subtask 8.2.3. Creation of original content (ESCI, LASEA, CEIT, MULTITEL, VISUM, FAGOR, RESCOLL, CIC)
A variety of editorial and visual content will be developed to share on LASER4SURF channels, media multipliers and social media. By focusing on the message, not the medium we will help create a credible, sustainable interest for largescale adoption of LASER4SURF results and solutions. The following content (see detail in section 2.3) will be produced: - Interviews: Every year, three to five people internal and external to the project consortium (technical experts, end users and stakeholders) will be interviewed and a total of 12 interviews will be produced.

- Original journalistic articles: As LASER4SURF gains momentum, a series of six original journalistic articles willprofile the skills, experiences, credibility and performance of the demonstration use cases and project in more detail. - Info-graphics: six info-graphics demonstrating key concepts, interoperability issues and results will be prepared. These are highly useful in many scenarios, including twitter, presentations, postcard flyers and posters. An initial series of three info-graphics focused on key concepts of the project will be defined to help attract audiences and build on the first social media actions. A further three will be produced towards the end of the project to profile key results.
- Brochure: A project's brochure will be designed in M6 in order to provide additional information on printed materialfor fairs and conferences.
- Audiovisual products: Two audio-visual products will be produced in professional HD broadcast standard to promotethe project. First, a short, concise and creative introductory video using animation will be developed within the first four months and will be used to capture and keep the attention of targeted audiences to stay informed about the project. The second one (produced at the end of the project) will show the results of the project, becoming an important tool also for the exploitation of project's results.
- Peer-reviewed scientific papers and poster presentation: different peer-reviewed scientific papers will be preparedduring the project lifetime to promote project results, as well as technical papers or poster presentations (see section 2.3 for further details). ESCI will support the LASER4SURF partners in this task, in particular in the creation of compelling poster presentations, and will use their online channels to disseminate these publications to the relevant target groups. Open access channels and platforms such as Researchgate will be used to support transparency and share insight.

Outcomes: Production of different editorial and visual content to share on LASER4SURF channels, media multipliers and social media.

Subtask 8.2.4 Online dissemination channels (ESCI, LASEA, CEIT, MULTITEL, VISUM, FAGOR, RESCOLL, CIC) This subtask is devoted to plan and implement all the activities related to online dissemination channels:
- Project website: this subtask will include the design, development and continuous update of the project website. It willbe regularly updated with information about the advance of the project, articles, project highlights and deliverables. As describe in section 2.2.2, to facilitate collaboration among partners, a protected space will feature appropriate working documents, reporting and non-public deliverables.
- Social network profile management: The project will have profiles in professional and social online networks suchas LinkedIn (to feed with project news and developments and targets invited to follow), twitter (to distribute original content and enhance and amplify presence before, during and after workshops and other events) and youtube (to host videos about the project). Besides LASER4SURF editorial content will be pushed to a number of websites for science, technology and specialist media. LASER4SURF will remain open to using any appropriate social media network or tool to meet the right target audience. Particular attention will be paid to social broadcasting tools like Periscope, Facebook
Live or USTREAM.

Outcomes: Project website and social media activities are ongoing.

Subtask 8.2.5 Print media and offline dissemination channels (ESCI, LASEA, CEIT, MULTITEL, VISUM, FAGOR, RESCOLL, CIC)
This subtask comprises the development of printed communication and dissemination materials to be used to support face to face meetings, at events, workshops and conferences and to extend exposure to LASER4SURF actions through poster sessions and other static displays. In this regard, brochures and info-graphics defined in subtask 8.2.3 will be developed for offline dissemination activities. Besides, a clear call to action for targeted audiences will be developed. Besides, to respond to the many opportunities for static display or visual support, an A1 poster with key facts and outputs of the project will be available for all consortium members.

Outcomes: Printed communication and dissemination materials available.

Subtask 8.2.6 External Workshops (ESCI, LASEA, CEIT, MULTITEL, VISUM, FAGOR, RESCOLL, CIC)
The workshops will be organised and managed by LASEA, and supported by all the other partners.
To ensure optimum outreach, at least three events will be organised back to back or in the context of other larger relevant sector-specific events such as LASYS, World of Photonics or the AILU workshops in order to attract a high number of stakeholders.

Outcomes: Workshops related to issues covered in LASER4SURF organised.

Subtask 8.2.7 Final Conference (LASEA, CEIT, MULTITEL, VISUM, FAGOR, RESCOLL, CIC, ESCI)

In M36, a final event will be organised at the end of the project to present all the final results of LASER4SURF. It will be open to a wider audience including stakeholders, public authorities, potential end-users and European institutions and will be organised in conjunction with the demonstration activity at the end of the project.

Outcomes: Final conference about LASER4SURF result open to all the targeted groups of LASER4SURF (Industry, scientific and academic community, public bodies and general public).

Subtask 8.2.8 Monitoring (ESCI, LASEA)
ESCI will define the relevant tools in the Communication Master Plan to monitor the impact of LASER4SURF's communication and dissemination activities. As online communication takes an important role in this project, and as it is such a fast moving environment, ESCI will study the impact of the dissemination measures every six months and review its dissemination success and adjust it if needed in M20.

Outcomes: Disseminations activities are analysed.

Participation per Partner

Partner number and short name	WP8 effort
1 - Ceit-IK4	
2 - LASEA	
3 - MULTITEL	
4 - VISUM	
IRIS	
5 - CICe	
CEGASA	
6 - FAGOR	
7 - RESCOLL	
8 - ESCI	
Total	

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D8.1	Business models and exploitation plans	2 - LASEA	Report	Confidential, only for members of the consortium (including the Commission Services)	12

D8.2	IPR management	1 - Ceit-IK4	Report	Confidential, only for members of the consortium (including the Commission Services)	36
D8.3	Communication Master Plan	8 - ESCI	Report	Confidential, only for members of the consortium (including the Commission Services)	3
D8.4	Project website and social media channels	8 - ESCI	Websites, patents filling, etc.	Public	4
D8.5	Introductory project video	8 - ESCI	Websites, patents filling, etc.	Public	4
D8.6	Brochure	8 - ESCI	Websites, patents filling, etc.	Public	6
D8.7	Final project video	8 - ESCI	Websites, patents filling, etc.	Public	36
D8.8	Interim report on dissemination and communication activities	8 - ESCI	Report	Confidential, only for members of the consortium (including the Commission Services)	20
D8.9	Final report on all dissemination and communication activities	8 - ESCI	Report	Confidential, only for members of the consortium (including the Commission Services)	36

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]

Description of deliverables

D8.1 Business models and exploitation plans (M12, M24, M36): This report will describe the overall exploitation strategy of the project results to be discussed and agreed during the project (Initial version and updates)
D8.2. IPR management (M36) - This deliverable will include the IPR management rules agreed upon the project.
D8.3. Communication Master Plan (M3)
D8.4. Project website and social media channels (M4)
D8.5. Introductory project video (M4)
D8.6. Brochure (M6)
D8.7. Final project video (M36)
D8.8. Interim report on dissemination and communication activities (M20)
D8.9. Final report on all dissemination and communication activities (M36)

D8.1 : Business models and exploitation plans [12]

This report will describe the overall exploitation strategy of the project results to be discussed and agreed during the Project. The deliverable will be updated in M24 and M36.

D8.2 : IPR management [36]

This deliverable will include the IPR management rules agreed upon the project.

D8.3 : Communication Master Plan [3]

This report will outline the communication and dissemination activities planned by the LASER4SURF partners for the whole project duration.

D8.4 : Project website and social media channels [4]

Project website and social media channels

D8.5 : Introductory project video [4]

Introductory video developed to capture and keep the attention of targeted audiences to stay informed about the project

D8.6 : Brochure [6]

Brochure to provide additional information on printed material for fairs and conferences.

D8.7 : Final project video [36]

Video with the results of the project.

D8.8 : Interim report on dissemination and communication activities [20]

This deliverable will detail the dissemination and communication activities carried out.

D8.9 : Final report on all dissemination and communication activities [36]

This report will summarized the dissemination and communication activities carried out.

Schedule of relevant Milestones

Milestone number[18]	**Milestone title**	**Lead beneficiary**	**Due Date (in months)**	**Means of verification**
MS8	Presentation of project website	8 - ESCI	3	Operational website

Work package number [9]	WP9	**Lead beneficiary [10]**	1 - Ceit-IK4
Work package title	Ethics requirements		
Start month	1	**End month**	36

Objectives

The objective is to ensure compliance with the 'ethics requirements' set out in this work package.

Description of work and role of partners

WP9 - Ethics requirements [Months: 1-36]
Ceit-IK4
This work package sets out the 'ethics requirements' that the project must comply with.

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D9.1	EPQ - Requirement No. 1	1 - Ceit-IK4	Ethics	Confidential, only for members of the consortium (including the Commission Services)	6

Description of deliverables

The 'ethics requirements' that the project must comply with are included as deliverables in this work package.

D9.1 : EPQ - Requirement No. 1 [6]

The applicant must provide further information about the possible harm to the environment and staff caused by the research and state the measures that will be taken to mitigate the risks. The applicant must confirm that facilities authorisations have been obtained (e.g. security classification of laboratory), and are kept on file. The applicant must ensure that appropriate health and safety procedures conforming to relevant local/national guidelines/legislation are followed for staff involved in this project.

Schedule of relevant Milestones

Milestone number[18]	Milestone title	Lead beneficiary	Due Date (in months)	Means of verification

1.3.4. WT4 List of milestones

Milestone number[18]	Milestone title	WP number[9]	Lead beneficiary	Due Date (in months)[17]	Means of verification
MS1	Identification of the type of beam shaping to be used (dynamic or passive)	WP2	3 - MULTITEL	18	Conclusions about analysis of costs, easiness of integration and robustness

MS2	Optical module	WP2	3 - MULTITEL	24	Validation of the module by laboratory experimental test
MS3	In-line monitoring module	WP3	4 - VISUM	24	Validation of the module by comparison with external characterization methods.
MS4	Optimized and Validated Software platform for parameter selection	WP4	1 - Ceit-IK4	22	Functional validation prior to integration
MS5	Operational validation of the global prototype	WP5	2 - LASEA	27	Functional test validation prior to carried out use case functionalization.
MS6	Feasible surface functionalization of use cases	WP6	6 - FAGOR	36	Tests completed and approved related to each use case.
MS7	Potential replications sectors and new applications identified	WP7	1 - Ceit-IK4	36	Replication guides agreed and prepared
MS8	Presentation of project website	WP8	8 - ESCI	3	Operational website

1.3.5. WT5 Critical Implementation risks and mitigation actions

Risk number	Description of risk	WP Number	Proposed risk-mitigation measures
1	A partner leaves the Project. Level of likelihood: Low	WP1	The PMC will analyse two main options: 1) the substitution of the partner by another one of similar characteristics 2) the assumption and redistribution of tasks among the partners of the project.
2	Delays on deliverables and results not meeting project objectives. Level of likelihood: Medium	WP1, WP2, WP3, WP4, WP5, WP6, WP7, WP8	The defined coordination and management structures have established mechanisms to react against possible delays. A project management handbook will be issued in M3. Regular WP and technical meetings will be held to ensure that activities are streamlined and learnt lessons shared.
3	Changes on planning execution. Level of likelihood: Medium	WP1, WP2, WP3, WP4, WP5, WP6, WP7, WP8	The structure of the management of the project will allow changes and it is designed to permit adaptations and modifications to the project execution if this proves improvements for results' achievement.
4	Beam splitting and shaping results on a too cumbersome system, with too many spots to be deflected by the mirrors of the galvo-scanner and then too difficult to handle with one single scanner. Level of likelihood: Medium.	WP2, WP5	The alternative might be to use two or three scanners in parallel to be still able to use all the different beams and total power from the laser and then still guarantee a sufficient process throughput. This would imply more integration and programming work from LASEA/MULT.

5	Need to use steep TopHat beams to guarantee a sufficiently homogeneous process resulting in a limited depth of focus optical system, very sensitive to working distance variations. Level of likelihood: medium	WP2, WP5	An autofocus will have to be implemented on the scanner to correct the distance between the sample and the scanner and guarantee always the same working distance.
6	Insufficient accuracy of the texturized surface monitoring system. Level of likelihood: Medium.	WP3	Other optical techniques will be considered to optimize the core device. The mechanical sampling system would also be equipped with stabilization system to reduce the noise affecting the measurements.
7	The process parameters selected by the SW tool give rise to LIPSS that differ from the target ones. Level of likelihood: Medium.	WP3, WP4, WP5	A feedback from the in-line monitoring system will be used to optimize and validate the selection of the parameters
8	The modified products do not fulfil the required operation. Level of likelihood: Medium.	WP5, WP6	Modifications will be made in the whole system and/or required modules
9	There are no suitable commercial electrode materials compatible with	WP4, WP6	Alternative non-commercial materials synthesized in-house will be chosen. If needed, a specialist company for materials scale-up will be contacted.

Risk number	Description of risk	WP Number	Proposed risk-mitigation measures
	the selected structuration pattern. Level of Likelihood: Medium.		
10	Wrong definition of the interfaces of the sub elements of the laser system. Level of likelihood: Medium.	WP5	To assure a good communication and collaboration between to the partners involved in the development of the different modules that have to be integrated later on between them.
11	The impossibility to reproduce the results obtained at other labs with the laser system integrating all modules.Level of Likelihood: Medium.	WP5	To collaborate in the DOE and preliminary trials testing first the reproducibility of those preliminary results.
12	Partners do not agree on the IPR of the results of the Project. Level of Likelihood: Low.	WP8	An exploitation plan will be developed within the first steps of the project, identifying the expected results (foreground), and who will be the owner. In addition, a Consortium Agreement will be signed by all partners before the project starts, establishing the basic rules for the management of the IPR issues regarding project results.
13	Partners are not committed to dissemination and communication actions. Level of likelihood: Low.	WP8	Regular 1:1 phone calls or audio conferences with project partners to explain and motivate the need to communicate

14	A partner is reluctant to communicate about the project, as they fear to unveil information to competitors. Level of likelihood: Medium.	WP8	An experienced science communicator would meet with this partner to help to define methods and terms how it is still possible to communicate effectively on the project without revealing any sensitive details.
15	Low interest in the workshops and the final conference. Level of likelihood: Medium.	WP8	Hook on other events and send out very target invitation in sufficient numbers.

1.3.6. WT6 Summary of project effort in person-months

1.3.7. WT7 Tentative schedule of project reviews

Review number [19]	Tentative timing	Planned venue of review	Comments, if any
RV1	18	Brussels	to be confirmed during the project
RV2	36	Brussels	to be confirmed during the project

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1. Project number

The project number has been assigned by the Commission as the unique identifier for your project. It cannot be changed. The project number **should appear on each page of the grant agreement preparation documents (part A and part B)** to prevent errors during its handling.

2. Project acronym

Use the project acronym as given in the submitted proposal. It can generally not be changed. The same acronym **should appear on each page of the grant agreement preparation documents (part A and part B)** to prevent errors during its handling.

3. Project title

Use the title (preferably no longer than 200 characters) as indicated in the submitted proposal. Minor corrections are possible if agreed during the preparation of the grant agreement.

4. Starting date

Unless a specific (fixed) starting date is duly justified and agreed upon during the preparation of the Grant Agreement, the project will start on the first day of the month following the entry into force of the Grant Agreement (NB : entry into force = signature by the Commission). Please note that if a fixed starting date is used, you will be required to provide a written justification.

5. Duration

Insert the duration of the project in full months.

6. Call (part) identifier

The Call (part) identifier is the reference number given in the call or part of the call you were addressing, as indicated in the publication of the call in the Official Journal of the European Union. You have to use the identifier given by the Commission in the letter inviting to prepare the grant agreement.

7. Abstract

8. Project Entry Month

The month at which the participant joined the consortium, month 1 marking the start date of the project, and all other start dates being relative to this start date.

9. Work Package number

Work package number: WP1, WP2, WP3, ..., WPn

10. Lead beneficiary

This must be one of the beneficiaries in the grant (not a third party) - Number of the beneficiary leading the work in this work package

11. Person-months per work package

The total number of person-months allocated to each work package.

12. Start month

Relative start date for the work in the specific work packages, month 1 marking the start date of the project, and all other start dates being relative to this start date.

13. End month

Relative end date, month 1 marking the start date of the project, and all end dates being relative to this start date.

14. Deliverable number

Deliverable numbers: D1 - Dn

15. Type

Please indicate the type of the deliverable using one of the following codes:

R Document, report
DEM Demonstrator, pilot, prototype DEC Websites, patent fillings, videos, etc.
OTHER
ETHICS Ethics requirement
ORDP Open Research Data Pilot

16. Dissemination level

Please indicate the dissemination level using one of the following codes:
PU Public
CO Confidential, only for members of the consortium (including the Commission Services)
EU-RES Classified Information: RESTREINT UE (Commission Decision 2005/444/EC)
EU-CON Classified Information: CONFIDENTIEL UE (Commission Decision 2005/444/EC) EU-SEC Classified Information: SECRET UE (Commission Decision 2005/444/EC)

17. Delivery date for Deliverable

Month in which the deliverables will be available, month 1 marking the start date of the project, and all delivery dates being relative to this start date.

18. Milestone number

Milestone number:MS1, MS2, ..., MSn

19. Review number

Review number: RV1, RV2, ..., RVn

20. Installation Number

Number progressively the installations of a same infrastructure. An installation is a part of an infrastructure that could be used independently from the rest.

21. Installation country

Code of the country where the installation is located or IO if the access provider (the beneficiary or linked third party) is an international organization, an ERIC or a similar legal entity.

22. Type of access

VA if virtual access,
TA-uc if trans-national access with access costs declared on the basis of unit cost,
TA-ac if trans-national access with access costs declared as actual costs, and
TA-cb if trans-national access with access costs declared as a combination of actual costs and costs on the basis of unit cost.

23. Access costs

Cost of the access provided under the project. For virtual access fill only the second column. For trans-national access fill one of the two columns or both according to the way access costs are declared. Trans-national access costs on the basis of unit

4. MEMBERS OF THE CONSORTIUM

4.1 PARTICIPANTS

4.1.1 ASOCIACIÓN CENTRO TECNOLÓGICO CEIT-IK4 (CEIT-IK4)

http://ceit.es/en/

4.1.1.1 DESCRIPTION OF THE LEGAL ENTITY AND ITS MAIN TASKS

Ceit-IK4 is a private multidisciplinary non-profit RTO closely connected to TECNUN, the Faculty of Engineering of the University of Navarra (Spain). Its mission is to provide the industry with services through the development of

technical research projects and to form young researchers and PhD students. Ceit-IK4 promotes excellence in applied research by publishing non confidential results and participating in scientific and technical forums. This RTO also has a comprehensive doctoral teaching program in the industrial area.

Ceit-IK4 has a staff of 249 employees and 51 PhD students and an annual budget over 15,3 M€. It consists of three vertical divisions (Materials and Manufacturing, Transport and Energy, Water and Health) and a fourth additional horizontal division (ICT).

Ceit-IK4 has been involved in 15 FP6 projects (having the role of coordinator in 2 of them), in 32 FP7 projects (having the role of coordinator in 10 of them) and is currently involved in 10 H2020 projects (being the coordinator in one of them).

Additionally, Ceit-IK4 is member of the IK4 Alliance, which covers a total of 1,400 people and has a turnover of around 100 M€. The IK4 Alliance was formed in 2005 in line with a federal model, whereby its members share strategies and combine capacities without giving up their sovereignty. Through a shared strategy, the sum of the capacities of the nine RTOs provides the alliance with the flexibility it needs to adapt to the characteristics of any company. Today, IK4 is a benchmark on the European scene and is among the continent's premier private, scientific and technological corporations. Attention should be drawn to IK4's leading role in the European Union's 7th Framework Programme, in which it has participated in over 200 research projects and led 63 of them.

CEIT-IK4 will participate in LASER4SURF project through the Additive Manufacturing and Laser Group of the Materials and Manufacturing division. The activity of the Additive Manufacturing and Laser Group focuses, among other on:

- the use of laser systems to process the material surface for a range of applications, including solar cells, semiconductor lasers and biomaterials.
- The development of optical of free space or integrated optical sensors using waveguides and diodes.

The Data Analysis and Information Management of the Information and Communications Technologies Division of CEIT will participate as well on the project, dedicated to transforming data into valuable information through: data processing, statistical analysis, determination of probabilistic algorithms, etc…

CEIT-IK4 will be involved in several WPs and tasks of the LASER4SURF project. From the most technical point of view, Ceit-IK4 will lead WP4- Modelling for tools for LIPSS processes and parameter selection. The main tasks within this WP will rely on the development of a software platform that will enable industrial end-users to select the process parameters that give rise to a desired functionality, without a need to deeply understand the LIPSS generation process. This duty is absolutely aligned with the expertise and experience of Ceit-IK4, as can be seen in the following sections.

In line with this WP4, the team will also be involved in:

- WP2- Design and development of a prototype for controlled LIPSS. The expertise of Ceit-ik4 in the development of LIPSS processes for advanced functionalities will be determinant in the definition of the requirements of the prototype
- WP3- Design and development of an in-line monitoring unit.
- WP5- Integration of the developed tools. As this WP will cover the integration of the modelling tools developed in WP 4, Ceit will decisively contribute to define the interfaces between subsystems and to test and optimize the interoperability of all of them.

On the other hand, as Coordinator partner of the project, CEIT-IK4 will be in charge of the coordination and management of the whole project at its different levels: technical development, IPR management, regulatory issues,

dissemination of the results, etc. Additionally, Ceit-IK4 will lead WP7 Gathering experience and new scenario definition with the aim of maximising the future impact of the project results.

4.1.1.2 CV OF THE PERSONS

- **Dr. Ainara Rodriguez** (female) is a researcher in the Additive Manufacturing and Laser Group of CEIT. She received her degree in Telecom Engineering from the University of the Basque Country in 2006 and her PhD from the University of Navarra in 2010. Since 2006, her research interests have been focused on laser material processing, including ultrafast processes in materials and Laser Interference Lithography. In this last field, she took part in the European Project "Development of lithography technology for nanoscale structuring of materials using laser beam interference" (EU FP6 ICT), as a result of which 7 scientific papers, 14 conference papers and 3 patent applications were obtained. She is also a lecturer of Optoelectronics and Optical Communications, courses of the degree in Telecom Engineering in the School of Engineering of the University of Navarra. During the last 10 years, she has been involved in 12 national and international research projects covering both basic and applied research and is co-author in more than 40 peer-reviewed papers and communications in conferences. She is also co-applicant in 3 patents.
- **Dr. Santiago M Olaizola** (male) is a researcher in the Additive Manufacturing and Laser Group of CEIT. He received his degree in Engineering in 1996, and PhD in Engineeering in 2000 from the University of Navarra. Later he received his PhD in Physics from the University of Sheffield in 2004. Since 2003, he has lead the Photonics Team from the Unit of Micosystems. He has focused in the field of micro/nanofabrication using pulsed and ultrafast laser sources. In this area, he has developed a 4-beam interference tool for industrial laser processes and currently he is also involved in the development of robust surface micro/nanofabrication processes using ultrafast laser sources. He routinely works in close contact with local Basque and Spanish companies in applications such as metrology, industrial manufacturing (polishing, plastic injection, ...) and optical sensors. He has been involved in 25 national and international research projects covering both basic and applied research and is co-author in more than 100 peer-reviewed papers and communications in conferences. He is also co-applicant in 7 patents.
- **Dr. Mikel Gómez** (male) received his MSc in Electrical, Electronic and Control Engineering from Tecnun (School of Engineering of the University of Navarra) in 2003. He completed his final degree project at the R&D department of the Telecontrol Division of IKUSI. After holding different posts at several industrial companies in which he took part in several research projects, in 2011 he completed his MSc in Biomedical Engineering at the University of Navarra. In January 2014, he received his PhD in Engineering from the same institution through his work on the development and microfabrication of microsensors with integrated microfluidic structures. Currently he is a researcher in the Additive Manufacturing and Laser Group of CEIT and his research interest includes laser interference lithography and laser induced surface modifications for industrial applications.
- **Dr. Jesús Gutiérrez-Gutiérrez** (male) is an Associate Professor at Tecnun (University of Navarra) and a researcher at CEIT. He received his degree in Mathematics from the University of Granada, Spain, in 1999 and his Ph.D. in Electronics and Communications from the University of Navarra, Spain, in 2004. His research interests include Statistical Signal Processing, Information Theory, and distributed computation in Wireless Sensor Networks. He is author of more than 20 scientific publications in international journals and has participated in more than 10 research projects.
- **Dr. Xabier Insausti** (male) received his Telecommunication Engineering degree, MSc. in applied Engineering an PhD. in Telecommunications Engineering in 2009, 2011 and 2013 respectively from the University of Navarra. He studied at Massachusetts Institute of Technology (MIT) for the academic year 2008-2009, where he completed his Final project. He received this degree with the highest average mark of the class. In 2014 he received his MSc. in advanced Mathematics from the UNED. He is currently a lecturer at TECNUN (University of Navarra) and a researcher in Information Theory, Coding & Compression Schemes for Communication Systems and in distributed computation in Wireless Sensor Networks.

4.1.1.3 LIST OF RELEVANT PUBLICATIONS

- M. Martínez-Calderon, M. Manso-Silván, A. Rodríguez, M. Gómez-Aranzadi, J. P. García-Ruiz, S. M. Olaizola, R. J. Martín-Palma, "*Surface micro- and nano-texturing of stainless steel by femtosecond laser for the control of cell migration*", Scientific Reports 6, 36296 (2016)
- M Martínez-Calderon, A Rodríguez, A Dias-Ponte, MC Morant-Miñana, M Gómez-Aranzadi, SM Olaizola, "Femtosecond laser fabrication of highly hydrophobic stainless steel surface with hierarchical structures fabricated by combining ordered microstructures and LIPSS", Applied Surface Science 374, 81-89, 2016
- Ainara Rodríguez, Maria Carmen Morant-Miñana, Antonio Dias-Ponte, Miguel Martínez-Calderón, Mikel GómezAranzadi, Santiago M. Olaizola, "*Femtosecond Laser-Induced Periodic Surface Nanostructuring of Sputtered Platinum Thin Films*", Applied Surface Science 351, 135-139 (2015)
- A. Pan, A. Dias, M. Gomez-Aranzadi, S. M. Olaizola, A. Rodriguez, "*Formation of laser-induced periodic surface structures on niobium by femtosecond irradiation*", Journal of Applied Physics 115 173101 (2014)
- X. Insausti, F. Camaro, P. M. Crespo, B. Beferull-Lozano and J. Gutiérrez-Gutiérrez, "*Distributed Pseudo-Gossip Algorithm and Finite-Length Computational Codes for Efficient In-Network Subspace Projection*" in IEEE Journal of Selected Topics in Signal Processing, vol. 7, no. 2, pp. 163-174, April 2013.

4.1.1.4 LIST OF RELEVANT PREVIOUS PROJECTS

- DELILA: "Development of lithography technology for nanoscale structuring of materials using laser beam interference", EU FP6 IST (2006-2009)
- LASERHUESO: "Laser fabrication of Surface micro/nanopatterns for the optimization of osteocells regeneration potential", CYCIT 2014 (Spanish Science Foundation)
- ECOGRAB: Design of a prototype for industrial engraving of diffraction/phase gratings for encoders applications using lasers (Spanish Science Foundation 2016-2018)
- Foundations and Methodologies for Future Communication and Sensor Networks (COMONSENS), Spanish Ministry of Science and Innovation - CONSOLIDER Project (Top Excellence Programme across all disciplines in Spain), PI: P. M. Crespo. Duration: 15/12/2008 - 14/12/2013.
- Radio Access Techniques for Heterogeneous Wireless Networks (RACHEL), Spanish Ministry of Economy and Competitiveness, PI: P. M. Crespo and J. Gutiérrez-Gutiérrez. Duration: 01/01/2014 - 31/12/2016.

4.1.1.5 DESCRIPTION OF ANY SIGNIFICANT INFRASTRUCTURE

CeitT-IK4 owns an ultrafast laser micromachining station for material processing that includes a Ti: Sapphire laser system consisting of a mode-locked oscillator and a regenerative amplifier which is used to generate 130 fs pulses at a central wavelength of 800 nm, with a 1 kHz repetition rate. The laser beam is focused on the samples using microscope objectives. This objective together with a CCD camera is used for online monitoring of the structuring process. A three dimensional translational stage moved the sample under the beam with different scan speeds allowing a variable overlap of pulses per spot. This setup has been used for surface functionalization of materials in order to produce superhydrophobic surfaces, decorative effects and optical devices.

Additionally, Ceit-IK4 counts on the equipment and facilities for the characterization of micro and nanostructures as well as for the characterization of specific functionalities. Some of the most relevant equipments are:

- SEM scanning electron microscope for the topographic characterisation of the patterned surfaces ▪ AFM Atomic Force Microscope for the characterization of the depth of the generated nanopatterns.
- Optical characterization lab for optical properties determination (reflectivity, absorbance, diffraction)
- Focused Ion Beam (FIB) and Transmission Electron Microscope (TEM) for detailed microstructural characterization.
- Built-in setup for Water Contact Angle measurements
- Climatic chamber for the characterization of the anticondensation behaviour of surfaces

4.1.2 LASER ENGINEERING APPLICATIONS SA (LASEA)

www.lasea.eu

4.1.2.1 DESCRIPTION OF THE LEGAL ENTITY AND ITS MAIN TASKS

Created in 1999, LASEA is a high-tech Belgian SME which has a unique expertise in the design, manufacture and use of laser systems for industrial purposes. More specifically, LASEA offers cutting edge laser solutions capable of responding to the needs of companies whose production lines include plastic welding, marking, and surface treatments.

Its laser products and services are: expertise and prototype development, OEM systems, laser workstations, turnkey machines. Equipped with 18 laser sources (CO2, high power diode lasers, diode pumped solid state lasers (DPSS) and fiber laser in cw, ns, ps and fs ranges), robots (6-axis and XYZC1C2 tables), as well as various measurements systems (pyrometer, power-meter and vision systems). LASEA is able to develop and validate its new process and systems before industrial integration.

LASEA is based on the Liège Science Park allowing it to have strong partnership with regional Universities and the Space Centre. With presently 40 people, LASEA has manufactured laser production lines all over the world (Europe, US, Australia ...). Certified ISO9001:2008, LASEA has 3 patents in the field of laser processing. For example, LASEA has developed since 1999 a method for locally removing a coat applied on a translucent or transparent substrate (patent EP1263538).

LASEA has already worked in several European Projects like FP6 Naginels, FP7 Sfera, FP7 PMJoin, Era-NET+ Plasma, Era-NET+ Sinead and regional projects Ecolas OLED, CZTS and SIMULAS. Currently LASEA is working in FP7 LASHARE, H2020 HIPERDIAS, H2020 POLAROLL, EUROSTARS HYIBIRD, MANUNET DECOAT and regional projects MICROLAB, GLASSLAS-3D, IMPLANTS-3D and MICROLAS.

LASEA's activity consists in implementing laser solutions in its laser station range. This range of stations is dedicated to particular applications according to the will of the customer: micromachining, texturing, plastic welding, decoating, engraving…. Therefore, the design and development of these stations is fitted to the special solution developed by LASEA's engineers for the customer's application.

LASEA has years of experience in laser processing including laser plastic welding, marking and micromachining. Therefore, LASEA will play a key role in the development of LASER4SURF project, contributing in the development of the Laser Texturing prototype mainly from the engineering aspects of the technology.

4.1.2.2 CV OF THE PERSONS

- **Ir. Axel Kupisiewicz** (m): founded LASEA in 1999. Chief Executive and Managing Director of LASEA. 17 years of experience in laser processing including laser plastic welding, marking and surface treatment. Technical manager of two European Projects and 3 regional R&D projects. Management of the company that is already exploiting, among other, the results of previous developed international projects.
- **Dr. Jose Antonio Ramos de Campos** (m) joined LASEA in 2013. Manager of the R&D department of LASEA. Broad experience of more than 25 years on lasers materials processing (cutting, welding, marking and micro-marking,

micromachining, cladding, etc.). Participation in various previous research projects, for example: STREP – FANTASIA (Flexible and Near-Net-Shape Manufacturing Chains and Repair Techniques for Complex Shaped Aero Engine Parts), CRAFT - CUSTOCER (Mass Customisation of Ceramic and Glass Decoration: a Contribution to the Future Manufacturing Industries), CRAFT – InART (Innovative Laser Based System and Technologies for In-Situ Cleaning of Paiting Artworks), amongst others.

- **Dr. David Bruneel** (m) joined LASEA in 2015. Member of the R&D staff of LASEA. A good experience of lasers processing (cutting, drilling, milling, marking, trepanning, etc.) and system development with different types of lasers (fiber lasers, ns, ps, and fs, using different wavelengths). Experience of several years in the Laboratoire Hubert Curien and Fraunhofer IBMT in developing USP process optimisation and systems' setups. 3 years of experience involved in previous R&D projects targeted at valuable commercially future applications SLIDE (Savings at Lubricated Interfaces Emission), OLED, CZTS (CZTS based photovoltaic solar cells).

4.1.2.3 LIST OF RELEVANT PUBLICATIONS Relevant publications:

- M. Alvarez, A. Henrottin, J.A. Ramos de Campos, P. Guaino, B. Heusdens, "Ablation de couches minces par impulsions (ultra-) courtes pour la production de composants photoniques", Journées Nationales des Procédés Laser pour l'Industrie, 2014.
- Henrottin, M. Alvarez, J.A. Ramos-de-Campos, F.J. Navas, M.A. Jimenez, G. Morales-Cid, "Microtexturation by laser for hybrid metal polymer joining", Journées Nationales des Procédés Laser pour l'Industrie, 2014.
- "Ultra-short pulse fiber laser parameters optimization for CdTe thin film solar cells processing and fiber laser design", Fiber Laser Applications, OSA, 2011.
- E. Mottay; A. Kupisiewicz; G. Detroux; X. Costet; A. Simmons; U. Vivarelli; P. Lemaire; J. Lopez, "Industrial ultrafast internal engraving laser system for anti-counterfeiting applications", Proc. SPIE 6881, 2008.
- B. Neuenschwander, B. Jaeggi, M. Schmid, V. Rouffiange and P-E. Martin (2012), "Optimization of the volume ablation rate for metals at different laser pulse-durations from ps to fs"; Proc. SPIE 8243, Laser Applications in Microelectronic and Optoelectronic Manufacturing, 824307

Relevant patents:

- Kupisiewicz, Axel: "Method for locally removing a coat applied on a translucent or transparent substrate". Wallonia Space Logistics December 11, 2002: EP1263538-A1.
- Kupisiewicz, Axel, Mottay Eric: "Method for internal laser marking". Trackinside Société Anonyme, Kupisiewicz Axel, Mottay Eric March 29, 2007: WO/2007/033445.
- Kupisiewicz, Axel, Mottay Eric: "Method for internal laser marking in transparent materials and device for implementing said method". Laser Engineering Applic, Ks Tech, Amplitude Systems, Costet S, Solos Identificazione E Protez, Total Brand Security. 2008: EP1926603-A1.

4.1.2.4 LIST OF RELEVANT PREVIOUS PROJECTS

Project acronym/description/website	Scope	End Year
PMJOIN: Development of a direct laser joining of hybrid plastic-metal components for industrial applications http://www.pmjoin.eu/	EU-FP7	2016
LASHARE: Laser equipment assessment for high impact innovation in the manufacturing European industry https://www.lashare.eu/	EU-FP7	2017

HYBRID: Development of an innovative laser system, introduced in a laser workstation prototype, able to join dissimilar materials of different organic and inorganic chemical nature like thermoplastics (or composites) and metals. https://www.eurostars-eureka.eu/project/id/9775	EUROSTARS	2019
HIPERDIAS: High throughput laser processing of diamond and silicon http://www.hiperdias.eu/	EU-H2020	2019
PoLaRoll: Polygon scanner based ultra-short pulse laser processing in roll-to-roll manufacturing http://cordis.europa.eu/project/rcn/205597_es.html	EU-H2020	2019

4.1.2.5 DESCRIPTION OF ANY SIGNIFICANT INFRASTRUCTURE

LASEA is equipped with 18 laser sources (CO2, high power diode lasers, diode pumped solid state lasers (DPSS) and fiber laser in cw, ns, ps and fs ranges), robots (6-axis and XYZC1C2 tables), various measurements systems (microscopes, pyrometer, power-meter and vision systems). Some of those resources are described below.

- Kyla software: LASEA develops software allowing the creation of tridimensional designs with an automatic definition of the laser drawing. Importation of dxf (2D) or stl (3D) files is also possible. The structures able to be created with the software can be very complex and may also allow the study of the laser parameters of different types of materials by the use of matrices of parameters. Its ease of use and the fact that all the laser (power, energy, repetition rate,…) and the scanner (trajectories, speed,…) parameters can be managed in one unique software allows a high productivity in terms of tests or in terms of production.
- Femtosecond experimental setups: for high precision applications, LASEA has the possibility to use an innovative technology: a laser which has pulse duration near of 400 femtoseconds (fs). The interaction of these types of beams with almost every type of material (even transparent to the wavelength of the laser) is so short that the ablation is almost athermal. This allows very precise, selective and very nice quality micro-machining. This technology may be used with a lot of different configurations and focalization systems to reach different processes.
- Confocal microscope: LASEA dispose of a high precision confocal for analyzing the laser-material interaction at the basically point of view of the pulses, lines drawn or at the higher scale of the channels and surfaces. This is particularly interesting in the case of the analysis of the micro-structuration of materials. Its resolution can go up to 10nm in the z axis and its resolution in the x and y axis may reach 0.12µm.
- Low tracking error scanners: The movement of the laser beam on a piece is obtained by passing through a scanner head, which consist in two motors on which are placed coated mirrors. The reaction time of a servo system is never zero (because of inertia) and the real positioning of the mirrors of the scanner has always a delay compared to the order. This delay in the case of the scanners dedicated to the applications of laser marking in the market is often included between 200 and 300µs, which doesn't allow a precise positioning at high speeds. LASEA uses a scanner technology called "Lorentz force" scanners: the performances are increased in the objective to reduce the tracking error to 70µs, which allows speeds up to some meters per seconds, depending of the application (micro-machining, ablation…). Moreover, these scanners have a low current consumption, resulting in a low heating and a low thermal drift: (on 8h) 24µrad compared to most scanners of the market (+/-600µrad). This increases a lot the reproducibility and the precision for example in the case of laser micromachining.
- Precision XYZ axis: the field size of the scanners, depending principally of the focal distance and the type of focal lens which is used, is limited, most of the time in values near of 50x50mm². Axes allow extending this field. With the Kyla software it is possible to cut automatically a dxf file in scanner zones linked one with each other with a movement of the axes. A synchronisation between the two types of the movements at the same time is also possible. In the x and y axes, a movement of 400mm is allowed at high speeds (up to 2000mm/s); in the z axis movement of 300mm is allowed. The resolution of positioning of both axes can be up to 1,5µm, what is perfect for laser micromachining, knowing that the "beam tool" itself, characterized among others by its diameter, may value about 10-15µm.

4.1.3 MULTITEL ASBL (MULTITEL)

www.multitel.be/photonics

4.1.3.1 DESCRIPTION OF THE LEGAL ENTITY AND ITS MAIN TASKS

MULTITEL is a non-profit organisation private research centre. The activities of MULTITEL include signal processing, wireless networks, railway certification, image processing and applied photonics.

The activities of MULTITEL include the following five R&D areas: The activities of MULTITEL include signal processing, wireless networks, railway certification, image processing and applied photonics.

Its activities include fibre lasers design and prototyping, laser micro-machining, optical sensors and terahertz imaging and spectroscopy.

The activities related to fibre lasers and micromachining include in particular the design and application of ultrafast lasers in various materials processing like thin films photovoltaics processing, surfaces texturization, glass refractive index modification. Other possible examples of applications are the glass processing (cutting, drilling, surface and embedded marking) or the polymer processing (particularly by resonant infrared ablation).

Due to its vast experience in ultrashort pulse lasers, in LASER4SURF project, MULTITEL will be addressing the beam shaping issues, in particular, through the leading of WP2. Besides, it will collaborate in the integration of the optical module into the processing unit developed in WP5.

4.1.3.2 CV OF THE PERSONS

- **Dr. Hernandez Yves** (Male): Dr. Hernandez joined the Applied Photonics Department of Multitel in May 2003. He previously worked with Alcatel Optronics in France as responsible of the product development, regarding the reliability of Semiconductor Laser Diodes. From 2003, Dr. Hernandez leads the Fiber Laser Group of the Applied Photonics Department. Since May 2013, Dr. Hernandez is Head of the Applied Photonics Department. He was coordinator of the FP7 Project CHARMING.
- **Dr. Jean-Bernard Lecourt**, (Male): Dr. Jean-Bernard Lecourt joined the Applied Photonics Department of Multitel in January 2007 after graduating from Université de Rouen (France) with a Ph.D. thesis on Femtosecond Lasers based on Saturable Absorbers (Etude et réalisation de lasers à fibre auto-impulsionnels à base d'absorbants saturables). At present time Dr. Lecourt is involved on R&D activities on design and prototyping of femtosecond fibre lasers and chirped-pulse amplifiers. Dr. Lecourt will take care of the activities related to ultrashort fibre lasers beam shaping and frequency conversion in WP2.
- **Mr. Julien Dupuy**, (Male): Ir. Julien Dupuy joined the Applied Photonics Department of Multitel in February 2014. He previously worked as an opto-electronics engineer in a french PME specialized in fiber optic system during 8 years. At present time Mr. Dupuy is in charge of the ultrafast laser micro-machining platform, for research and industrial purpose. He is also involved on R&D activities of fiber optics sensors, and prototyping of femtosecond

fibre lasers. Ir. Dupuy will take care of the validation of the optical beam shaping systems in the current micromachining MULTITEL platform before integration in LASEA system.

4.1.3.3 LIST OF RELEVANT PUBLICATIONS

- A. Bertrand, Y. Hernandez "Exploring new avenues for laser micromachining", SPIE Newsroom, 10 May 2016, DOI: 10.1117/2.1201603.006363
- Jean-Bernard Lecourt ; Simon Boivinet ; Anthony Bertrand ; Didier Lekime ; Yves Hernandez "High-energy picosecond hybrid fiber/crystal laser for thin films solar cells micromachining", Proc. SPIE 9513, High-Power, High-Energy, and High-Intensity Laser Technology II, 951303 (May 12, 2015); doi:10.1117/12.2180234
- S. Boivinet ; J.-B. Lecourt ; Y. Hernandez ; A. A. Fotiadi ; M. Wuilpart ; P. Mégret "Impact of chromatic dispersion and spectral filtering in an all-fiber mode-locked ytterbium laser", Proc. SPIE 9344, Fiber Lasers XII: Technology, Systems, and Applications, 934421 (March 4, 2015); doi:10.1117/12.2078865
- S. Boivinet, J.-B. Lecourt, Y. Hernandez, A. Fotiadi, P. Mégret "948 kHz repetition rate, picosecond pulse duration, all-PM 1.03 µm mode-locked fiber laser based on nonlinear polarization evolution", Photonics Europe 2014, Session 12: Mode-Locked Lasers, paper 9135-55
- Yves Hernandez, Charles Duterte, Erwin Lotter, Alessio Bosio, Marc Hueske, and Stefano Selleri "Picosecond fiber laser applied to solar cells scribing*, Anthony Bertrand"*, International Laser Applications Symposium (ILAS) 2013,

4.1.3.4 LIST OF RELEVANT PREVIOUS PROJECTS

Regarding micromachining, Multitel has worked in process optimization for solar cells in the FP7 projects ALPINE, IMPROV and LASHARE. In the CORNET project ADOPT, Multitel develops a machining system coupled with online monitoring. Multitel coordinated the INTERREG project ECOLASERFACT related to laser micromachining promotion in North West Europe.

- **ADOPT** "ADaptive laser micro machining of multi-layer surfaces with inline OPTical metrology" (CORNET, convention n°1510569)
- **LASHARE** "Laser equipment ASsessment for High impAct innovation in the manufactuRing European industry" (FP7-2013-NMP-ICT-FOF, Grant Agreement n°609046, www.lashare.eu)
- **ECOLASERFACT** "ECO-efficient LASER technology for FACTories of the future" (Interreg IV B NWE, www.ecolaserfact.eu)

  ECOLASERFACT aims to transfer to North-West Europe based SMEs the most recently developed laser micromanufacturing processes for fast, flexible, high-precision, reduced cost and ecological friendly manufacture of goods. ECOLASERFACT aims to federate actors involved in laser micromanufacturing leading to an efficient transnational Network or Expertise. With the help and services provided by the Network of Expertise, the consortium supports SMEs with the latest laser manufacturing technologies to produce highly functional and innovative products by an easy and cost effective access to on-demand processing equipments.
- **ALPINE** "Advanced Lasers for Photovoltaic INdustrial processing Enhancement" (FP7-NMP-2008-LARGE-2, Grant Agreement n° 229231, www.project-alpine.eu)
- **IMPROV** "Innovative Mid-infrared high Power source for resonant ablation of Organic based photovoltaic devices" (FP7-ICT-2009-5, Grant Agreement n°257894, www.fp7project-improv.eu).

4.1.3.5 DESCRIPTION OF ANY SIGNIFICANT INFRASTRUCTURE

Multitel laser micromachining platform provides infrastructure and technical equipment to serve the objectives of the project. It offers the following:

- **Facilities**:
  - Two clean rooms
  - Flexible micromachining platform ▪ **Lasers available:**
  - Amplitude S Pulse/Central wavelength 1030nm/Pulse duration 540fs-10ps/Av. Power 3.5W/Pulse energy 35µJ/Rep rate100kHz max
  - Amplitude S Pulse/1030nm/250ps/3.5W/150µJ/25kHz max

- Amplitude S Pulse/515nm/540fs/1.5W/15µJ/100kHz max
- Amplitude S Pulse/515nm/250ps/2W/80µJ/25kHz max
- Amplitude S Pulse/343nm/250ps/500mW/20µJ/25kHz max
- Amplitude S Pulse/257nm/540fs/300mW/3µJ/100kHz max
- Amplitude S Pulse/257nm/250ps/300mW/12µJ/25kHz max
- IPG YLPN-1-4x200/1064nm/8-200ns/20W/1mJ/20kHz-100kHz
- Multitel 30ps/1064 nm& 532 nm/ 100 µJ / 5W
- Multitel 1 ps/1030 nm/ 100µJ – burst mode laser
- Multitel ns/1040nm/40ns/20W/0.25mJ/80kHz
- Multitel CW/1064nm/CW/100W

- **Scanners/translation stages available:**
  - Three galvanometric heads with mirrors at 355nm, 515nm, and 1030nm
  - Adapted f-theta lenses and microscope objectives
  - Translation stages Aerotech (max speed 3m/s, Travel X 500mm/Y 200mm/Z 35mm, XY resolution 10µm, Z resolution 5µm)
- **Means of characterization:**
  - Microscope (objectives X5 X10 X20 X100 X150)
  - Optical profilometer (XY resolution 5µm, Z resolution 100nm)
  - Access to Scanning Electron Microscope (SEM)

4.1.3.6 OTHER DOCUMENTS

4.1.4 VISUM LTD. (EX. IRIS ADVANCED ENGINEERGING LIMITED) - VISUM

www.seeingnewdata.com

4.1.4.1 DESCRIPTION OF THE LEGAL ENTITY AND ITS MAIN TASKS

Visum Ltd.(Ex. IRIS Advanced Engineerging Limited) (VISUM) www.seeingnewdata.com is an optical engineering company from Dublin that specializes in the development and commercialization of Near Infrared Spectroscopy (NIRS)-based portable and inline analyzers and measurement systems for process monitoring and quality control applications in the food, pharmaceutical and chemical applications. The company was established in 2014 and currently employs 9 staff from its facilities in Nexus UCD (technology park at the University College Dublin campus). In 2016, the company launched the VISUMPalm (a handheld NIR measurement instrument) and has secured first sales. An integration platform (VisumNIR) for the inline version is currently under development in cooperation with early adopter clients from the food processing industry (dairy, meat and olive oil) and pharmaceutical industries. VISUM Ltd. is an SME Champion-awarded a Phase 1 and a subsequent Phase 2 Grant under H2020's SME Instrument.

The optical engineering and smart systems teams at VISUM Ltd. have specialized expertise in sensor integration, optical engineering, chemometrics, data analytics, cloud infrastructures and data management systems.

VISUM will be responsible for the WP3 Design and development of an in-line monitoring unit, which addresses the development of the in-line quality control system of the laser texturing prototype. Moreover, will contribute on defining the optical technology suitable for nanometric surface features quality control. Developing a functional prototype suitable for industrial settings use and testing and validating the new nano-metrology prototype surface functionalities and performance. Furthermore, transversal contributions will be given to the consortium in terms of Exploitation and Dissemination aspects.

4.1.4.2 CV OF THE PERSONS

- **Dr. Timothy Kehoe** (Male) PhD in Engineering, University College Cork, Ireland- developed an IR photometric measurement system for thermal diffusivity in anisotropic materials. Masters in Optoelectronics, Queens University Belfast. Designed an Echelle spectrometer while based at Andor Technology Ltd. Worked at Tyndall National Institute, Cork, and Catalan Institute of Nanotechnology, Barcelona, developing diffractive optical and acousto-optical metrology techniques for nanotechnology applications, in particular for characterisation of nanoscale structures fabricated by nanoimprint lithography and directed self-assembly. Other research activities: quantification of ordering in self-assembled structures by image analysis techniques and optical sensors using hydrogel materials. Tim is the VISUMPalm Product Manager, as well as the Technical Lead at VISUM Ltd. and he brings vast optical integration and device instrumentation expertise.
- **Dr. Severine Philippe** (Female) graduated with a 1st class Hons. B.Sc. in Applied Sciences (Physics and Physics Technology) from Dublin Institute of Technology, Ireland, in 2002 and received her PhD in Physics from Trinity College Dublin in 2007, for the study of polarisation effects in semiconductor optical amplifiers (SOA). She went on to work on the development and characterization of low noise SOAs, as well as THz generation by multimode lasers and their detection using four wave mixing in SOA, both in the Semiconductor Photonics Group, Trinity College Dublin, and the School of Electronic Engineering, Dublin City University. She previously worked as a R&D technician for the plastics industry and in quality control for electronic appliances. She joined VISUM in 2014 as

an optical engineer and is currently involved in projects including various microscopy techniques, planar waveguide technologies and MIR spectroscopy.

- **Nikolina Mikulcic** (Female) User Interface Designer. Nikolina comes from web design and development background with more than 15 years of experience in this field and she is an expert on UI/UX design strategy and front-end development (CSS/HTML). She studied graphic design and she started to learn web design due to her deeply interested for digital world. Since joining VISUM in September 2016, she has been creating a new User Interface for the VISUM product lines. Prior to VISUM, she has an extensive experience in the online world, having worked in companies such as EuroArt93 – digital agency, BlueBOX IT, Biroval. She also was a founder of design studio In Love With Blue and she worked as Project Manager, UI/UX designer and front-end developer. She was leader numerous digital projects for different clients as Erste Bank, Unicredit Bank, Telenor Bank, Komb Bank, ZAMP etc.
- **Chris Dermody** (Male) Software Developer. Bachelor's degree in Product Design from Dublin Institute of Technology. Chris excels in creative problem-solving, using the latest technologies to transform ideas into real opportunities that provide exponential value to business operations. With a keen eye for user experience, Chris builds robust, responsive applications that stand the test of time. Since joining VISUM, Chris has been hard at work developing both front and backend features that push the VISUM product lines forward. Prior to VISUM, Chris enjoyed working across the full spectrum of business, founding startups and overseeing product development throughout the lifecycle of many successful businesses.
- **Dr. Idoia Martí Aluja** (female) is a Senior Scientist. She received her PhD in Chemistry at the Universitat Rovira i Virgili (Spain). Her thesis was focused on the establishment of novel analytical methods for the study of different processes by infrared spectroscopy and chemometrics techniques. After her thesis, she expanded her knowledge in spectroscopic techniques working as a Technician in the Spectroscopic and Kinetics Unit (SKU) at the Institut Català d'Investigació en Química (ICIQ, Spain). She was in charge of the calibration and maintenance of the instruments of the unit (UV-Vis, NIR, FTIR, FT-Raman, Circular Dichroism, Fluorescence spectra and polarimetry). Moreover, she worked as a PostDoct at the Institute of Photonic Sciences (ICFO, Spain), leading a project whose aim was to develop a new and novel infrared instrument based on a high power laser source with direct emission in the mid- to long-wave infrared for chemical microspectroscopy of biological samples. She was using chemometrics techniques to extract meaningful information from the recorded data. She also worked as Material Science Engineer at Zobele Spain, being in charge of the selection, testing and validation the most adequate raw materials, with particular focus on the interaction between chemical products (mainly perfumes, solvents and detergents) and materials.

4.1.4.3 LIST OF RELEVANT PUBLICATIONS

- Development and commercialisation of the VISUM instruments- NIR spectroscopy devices- in-pipe, handheld
- Publication in relation to Innovation Management: O. Mc Nerney, A. Stachowicz, P. Czuprynski, E. Philpott, D. Kolman, D. Martin, E. Herrmann (2010). "Strategic Guide to Successful Use and Dissemination of the Results of Research and Development Projects" 1st edition, Steinbeis-Edition Stuttgart 2010. ISBN 978-3-941417-27-4
- On 08/02/2016, **VISUM was awarded the Seal of Excellence form the European Commission**

4.1.4.4 LIST OF RELEVANT PREVIOUS PROJECTS

- **PREVIEW** "PREdictiVe system to recommend Injection mold sEtup in Wireless sensor networks". (Grant Agreement number: 636892).

  PREVIEW aims to develop cutting edge wireless technological solutions for injection moulding setup and process optimisation using recommended machine parameters to minimise manufacturing downtime. VISUM Ltd. is responsible for the development of the Data Acquisition System and in developing the Signal Conditioning, its integration with the ARM main board, as well as implementing a prototype.
- **Phase 1 SME Instrument**: An IR-spectroscopy-based platform for the rapid integration and customisation of affordable, real-time monitoring solutions for the food and pharmaceutical industry (Grant Agreement number: 684767).

- **Phase 2 SME Instrument**: Cloud-enabled IR spectroscopy platform for the rapid integration and customisation of affordable, real-time chemical monitoring solutions for food and pharmaceutical industries (Grant Agreement number: 731043)
- Development of a portable LIDAR device for the detection of pesticides for the University of Lerida in Spain (B2B contract)
- Development of a NIR device for at-line measurement of quality of oil in industrial frying equipment for bakery products (B2B contract).

4.1.4.5 DESCRIPTION OF ANY SIGNIFICANT INFRASTRUCTURE

VISUM has the following infrastructures that will be used in the project:

Spectroscopic Laboratory: equipped with several instruments, namely, an FTIR-ATR (Interspectrum), two UV-VIS (PerkinElmer and Ocean Optics), one Extended-SWIR (StellarNet) and one NIR (Mightex) spectrophotometers as well as a SWIR Hyperspectral imager (Specim-Xenics). The associated staff are trained and skilled in chemometrics modelling by using a commercial suite (SOLO+MIA from Eigenvector) as well as ad hoc non-commercial chemometric tools (Matlab and Python toolboxes).

Electronics, mechanical and optical workshops: equipped with up-to-date hardware and tools: signal generators, wide band oscilloscopes, PCB design [Altium] and building tools, development kits for programming microcontrollers, CNC-milling machines for electronic and mechanical prototyping, and CAD software [SolidWorks], among others. ZEEMAX software for the design of optical systems and design setup by means of advanced optical components and facilities (optical tables, windows, lenses, lasers and light sources and related “opticalware”).

4.1.5 CENTRO DE INVESTIGACIÓN COPERATIVA DE ENERGÍAS ALTERNATIVAS FUNDACIÓN (CICe)

www.cicenergigune.com

4.1.5.1 DESCRIPTION OF THE LEGAL ENTITY AND ITS MAIN TASKS

CICe is a non-profit energy research centre based in the Basque Country (Spain), funded in 2011. CICe's mission is to play a leading role on the international scientific stage focused on basic research in energy related materials oriented to storage applications, contributing to industrial competitiveness through: excellent and breakthrough research; transfer of technology and knowledge to industry; coordination of basque technology and research efforts in energy storage field. CICe plays an important role in the strong Basque Network of Science, Technology and Innovation (RVCTI). The centre has the backing of public institutions, administrations and of companies directly related to the energy sector. Its objectives are strongly aligned with the Basque Country RIS3 priorities and the Energibasque strategy, which identifies storage as the central key to solve the future energy needs.

In the project, besides all the administrative and dissemination and exploitation tasks, CICe will mainly participate in WP4 and WP6 with the following tasks:

- Participation in the elaboration of a database with LIPSS processes and resulting functionalities. The role of CICe will be focused on the processes associated to metals typically used in batteries. Moreover, CICe and CEGASA (third party) will participate in the definition of the pattern for battery current collector.
- Fabrication of Li-ion battery electrodes to test the capabilities of current collectors processed by LIPSS. CICe will elaborate the electrode formulation with the input provided by CEGASA, electrodes will be casted and assembled in half cells and full cells for subsequent electrochemical testing that will be used to determine the improvement obtained after the LIPSS processing. Finally, pouch cells will be assembled to prove the scalability of the process.
- Characterisation of the current collectors and electrodes. Advanced characterisation will be performed on cycled electrodes that will help us to determine the influence of the LIPSS processing in the electrochemical performance, lifetime and stability of the electrodes. CEGASA will provide valuable input when defining the testing protocols and experimental conditions.

4.1.5.2 CV OF THE PERSONS

- **Dr. Miguel Ángel Muñoz** (Male) PhD from Warwick University in 2005. He is now associate researcher at the Structure & Surface Analysis Group. His research focuses on the understanding of the mechanisms that drive surface and interface reactions in electric energy storage components. Miguel Ángel is an experienced scientist active in the area of surface/interface characterization (XPS, STM/AFM, AES), thin film and cluster growth under UHV conditions.
- **Mr. Aitor Villaverde** (Male). He is the Platform Manager for the Dry Room and Prototyping Facility at CICe. Aitor has over 10 years experience in industry (GUASCOR FOTON S.L., BATZ cooperative). He began his career in renewables energies as a technical electronic engineer in 2004, in Mexico where he collaborated in the design and installation of an electronic control system for energy storage in photovoltaic systems. In 2005 Aitor received clean room training in California and The Institute of Microelectronic Technology (TiM) at the San Mamés Industrial Engineering School.
- **Dr.Laida Otaegui**, (Female). PhD in Industrial Engineering from Tecnun, Universidad de Navarra. During her undergraduate period, her research was focused in the fabrication and electrochemical/microstructural characterization as well as their long-term stability at high temperature and corrosive environments of different high temperature energy conversion and storage systems such as Solid Oxide Fuel Cells (SOFC) and High

Temperature Metal-Air Batteries (HTMAB). She is now a post-doc researcher at the Device Engineering Group in which she is working in the development of Na-ion battery electrode formulations and battery prototypes.

4.1.5.3 LIST OF RELEVANT PUBLICATIONS

- Composition and Evolution of the Solid-Electrolyte Interphase in Na2Ti3O7 Electrodes for Na-Ion Batteries: XPS and Auger Parameter Analysis. M.A. Muñoz-Márquez et al. ACS Appl. Mater. Interfaces 7, 7801−7808 (2015). A combination of photoemission techniques was used to determine the evolution of the electrode-electrolyte interphase in Na-ion batteries which provided new insights on the stability and capacity fade of this type of electrodes.
- Direct observation of electronic conductivity transitions and solid electrolyte interphase stability of Na2Ti3O7 electrodes for Na-ion batteries. M. Zarrabeitia et al. J. Power Sources 330, 78-83 (2016). Impedance spectroscopy studies and photoemission techniques were combined to deliver an answer regarding the transport properties of promising anodes for Na-ion batteries.
- Local Structure and Stability of SEI in Graphite and ZFO Electrodes Probed by As K‑Edge Absorption Spectroscopy. S. J. Rezvani et al. J. Phys. Chem. C 120, 4287−4295 (2016). Use of X-ray absorption spectroscopy to characterize the evolution of the electrode-electrolyte interphase in Li-ion batteries.
- Effect of the electrolytic solvent and temperature on aluminium current collector stability: A case of sodium-ion battery cathode. L. Otaegui et al. J. Power Sources 297, 168−173 (2015). Differences on the Na-ion battery current collector stability related to the use of carbonated and ionic liquid solvents for NaFSI based electrolytes studied by means of electrochemical and SEM analysis.
- Performance and long term stability of a liquid-tin anode metal-air solid electrolyte battery prototype. L. Otaegui et al. Electrochimica Acta 214, 192-200 (2016). Development and electrochemical/structural characterization of a high temperature liquid-tin anode metal-air solid electrolyte battery using galvanostatic cycling, impedande spectroscopy and SEM/EDAX.

4.1.5.4 LIST OF RELEVANT PREVIOUS PROJECTS

- MAT4BAT “Advanced materials for batteries” (FP7-GC.NMP.2013-1, Grant Agreement No. 608931). Development of advanced electrode formulations screening a wide variety of binder and stabilizing agents.
- SIRBATT “Stable Interfaces for Rechargeable Batteries” (FP7-ENERGY-2013-7-3-3, Grant Agreement No. 608502). Influence of battery interfaces in the cycle life and safety of Li-ion batteries.
- LINABATT “Li-ion and Na-ion battery technologies for grid and transport applications.” (Retos Investigación MINECO, Grant No. ENE2013-44330-R). Improvement of energy density, safety and cycle life of Li-ion batteries and development of new materials.
- Energigune14 “Development of research activities in electrochemical and thermal energy storage.” (Proyectos de Investigación con Alto Potencial Industrial, Grant No. IE14-388). Development of materials and cells for Naion battery technology.
- BATNALAE “Design, fabrication and test of Na-ion batteries for emergency luminary applications.” (Proyectos de Investigación Estratégica, Grant No. KK-2016/00098). Development of Na-ion battery prototypes for luminary applications.

4.1.5.5 DESCRIPTION OF ANY SIGNIFICANT INFRASTRUCTURE The

following infrastructures will be used in the project:

- **Dry Room / Prototyping Facility**: The dry room at CIC energigune is a 55 m2 laboratory in which humidity and temperature conditions are continuously controlled. Humidity levels as low as -60 ºC dew point at a 21 ºC can be reached in this platform. Moreover, the laboratory is fully equipped with the machinery required for different levels of battery prototype manufacturing, either coin cells or pouch cells. Due to the moisture sensitivity of most of the battery components this constitutes the perfect environment to build the cells which will be developed in this project.
- **Surface Analysis Platform:** This platform integrates state of the arte equipment devoted for advances studies on surfaces and interfaces. A high resolution XPS with 2D detector will allow to perform photoemission

experiments at different X-ray energies while probing diferent surface regions. The system is coupled to a Raman spectrometer fitted in a electrochemical cell that enables us to carry out all characterisation under controlled conditions without exposing the battery components to air. This is a very important point since the majority of battery systems are moisture sensitive.

- **Electron Microscopy Platform:** The scanning electron microscope is also equipped with a transfer system that allows sample handling without air exposure. This technique will be used to study the surface morphology of the electrodes and, with a special sample holder, cross-sectional studies of the electrode-current collector interface.

4.1.6 FAGOR AUTOMATION S. COOP (FAGOR)

www.fagorautomation.com

4.1.6.1 DESCRIPTION OF THE LEGAL ENTITY AND ITS MAIN TASKS

FAGOR AUTOMATION is a company with great experience in the development and manufacturing of products for machine automation and control. It excels by its capability to develop software and the versatility of its productive process to offer solutions tailored to the needs of their customers. Historically, FAGOR AUTOMATION has focused their efforts on the machine-tool sector where we are one of the world leaders. Besides, its wide technical and sales network spread out in more than 50 countries through its own branch offices and distributors. FAGOR AUTOMATION career has been so successful thanks to our investing efforts in R&D+i (Research, Development and innovation) and in infrastructure that made it possible for us to tackle projects of high strategic relevance. An example of this, are the new facilities for photolithography to produce engraved glass that is used in our linear feedback systems.

Traditionally, the company has dedicated more than 10% of its turnover to R&D+i; it currently has 85 R&D engineers and 6 doctors dedicated to developing new automation and optics technologies integrated into the products developed by FAGOR AUTOMATION.

An important part of their product portfolio is the position measuring systems. Fagor Automation has been manufacturing high quality linear and rotary encoders using precision optical technology for more than 30 years. Over the years Fagor has created, developed and patented systems, components and technologies that allow us to offer best quality and features over the complete range of product utilizing innovative production methods.

In order to ensure quality and reliability in all its products Fagor Automation utilizes the most advanced technology and testing and manufacturing facilities. From centralized computer control temperature monitoring, cleanliness and relative humidity control, a must for the feedback system manufacturing process, to laboratories for climate, vibration and EMC testing to certify the designs.

Fagor Automation will be one of the end users that will validate the use of the developed laser texturing prototype, particularly, for the functionalization of optical encoders. The proposed technology will be an alternative to the present photolithography that Fagor is using. During this project Fagor Automation will contribute to stablish an adequate target for the technology.

4.1.6.2 CV OF THE PERSONS

- **Dr. Tomás Morlanes** (M): He is a Doctor in Physics (by Universidad Complutense de Madrid) specialized in the optics field. He has always developed his work in the Research & Development department of Fagor. Now he is the Optics Technology Manager in Fagor. He works in the field of optical linear and rotary encoders.
- **Mr. Haritz Anduaga**, (M): The academic profile of Haritz Anduaga a Master in Mechanical Engineering by Mondragón Unibertsitatea. He began their work in Fagor like technical in the Manufacturing department. Now he is the Industrialization Manager of Feedback Systems.

4.1.6.3 LIST OF RELEVANT PUBLICATIONS

In the following links, 2 examples of products that will be obtained using the proposed products can be found:
http://www.fagorautomation.com/en/product/feedback-systems/linear-and-angular-encoders-for-cnc-machinesand-high-accuracy-applications/incremental/serie-l/

http://www.fagorautomation.com/en/product/feedback-systems/open-linear-encoders/absolutos/

4.1.6.4 LIST OF RELEVANT PREVIOUS PROJECTS

The following project is connected to the subject of LASER4SURF:

- LASERHUESO: "Laser fabrication of Surface micro/nanopatterns for the optimization of osteocells regeneration potential", CYCIT 2014 (Spanish Science Foundation)

4.1.6.5 DESCRIPTION OF ANY SIGNIFICANT INFRASTRUCTURE

The following equipment will be used in the project:

Equipment for Product Qualification

- Climatic Room 1-60ºC
- 2 Humidity and temperature Chambers
- Induced interference Generator
- Electrostatic discharge generator
- Radioelectric perturbations generator/receiver
- GETEM camera for radioelectric perturbations susceptibility/emission measurements

Equipment for Functional tests of Products

- Laser precision stations for displacements measurements
- Confocal Microscope for thin film measurements (down to 15 nm)
- Digital oscilloscopes
- Optical Laboratory

4.1.6.6 OTHER DOCUMENTS

NO

4.1.7 RESCOLL TECHNOLOGY CENTRE OF MATERIALS – RESCOLL

https://rescoll.fr/rescoll/

4.1.7.1 DESCRIPTION OF THE LEGAL ENTITY AND ITS MAIN TASKS

RESCOLL is an innovative SME, best defined as a research company for materials, specialized in composites and polymers. RESCOLL's activities are industrial research and development of innovative products and processes. RESCOLL is conducting innovation research in the field of industrial applications of polymers (composites, resins,

adhesives, varnish, coatings, etc.) and related fields: bonded assemblies, materials fire behavior and environmentally friendly surface treatments

RESCOLL currently has 80 employees, including PhDs, engineers and technicians with sales of 10 M€ in 2016. RESCOLL does more than 25% of its sales in the medical industry. RESCOLL has been at the origin of more than 40 patents in the field of polymers and composites. RESCOLL has ISO 17025 and NADCAP accredited laboratories and is specialized in the fields of polymer formulation, vibroacoustic testing, mechanical characterization, nondestructive control, fire testing, thermo mechanical analysis, spectrometry and chromatography and surface analysis and accelerated ageing.

RESCOLL also participates in several European projects including project within the scope of FP6 and FP7. In addition, RESCOLL has participated in several Clean Sky and CORAC projects.

RESCOLL also has extensive experience in the development of and testing of medical devices and sol-gel coatings. As a result, RESCOLL possesses considerable expertise in the characterization of the electrical properties of such system, and will use this expertise in the characterization of the insulating materials.

Its technical skills, quality system and the scope of its material means (technologically advanced equipment) make RESCOLL one of the state-of-the-art laboratories in the field of materials testing on the European level.

4.1.7.2 CV OF THE PERSONS

- Ing. Florent Deliane: technical manager. He is Head of Surface Technology Materials business unit managing 6 researchers. Ing. Florent Deliane has more than 10 years experience in surface technology, coating and sol-gel process. He was involved in the management of several partnership project dedicated to optics, corrosion, tribology, adhesion, catalysis concerning application in aeronautics, medical device, transport, or energy. Patents and product commercialization have been carried out for some of these projects.
- Etienne Lecomte (male): commercial manager. Mr. Lecomte has over 14 years' experience dealing with products and services destined for the public market and industry. He currently manages a team of 3 people. He serves for the account manager for large industrial groups such as AIRBUS, SAFRAN, and AREVA.
- Mrs. Magali Clavé-Henry (female): marketing manager. Magali graduated with an engineering diploma and a specialization in Fine Chemistry from the National Institute of Applied Sciences (Rouen – France) in 2005. She then worked for 3 years in the R&D department of a chemical group based in The Netherlands on the development of resins and dispersions for adhesive applications. In 2009, she joined a consulting company specialized in innovation management. Her areas of expertise include collaborative project development, market intelligence and advising innovative companies. She will be joining Rescoll as of the 1st of June 2015 and will be in charge of conducting market research and supporting strategic planning.

4.1.7.3 LIST OF RELEVANT PUBLICATIONS

RESCOLL provide with services in medical implant design and manufacturing. Therefore, this knowledge can be very useful for the project, specially to evaluate the replication potential of LASER4SURF technology in different products and/or applications of RESCOLL expertise fields.

4.1.7.4 LIST OF RELEVANT PREVIOUS PROJECTS

- **H&M** is a French national program focused on the development of innovative coatings for hip and knee implants.
- **MED FAB LAB** is a French regional project (Aquitaine) for an additive manufacturing platform for medical devices and applications.

4.1.7.5 DESCRIPTION OF ANY SIGNIFICANT INFRASTRUCTURE

MED FAB LAB medical device additive manufacturing platform. It focuses on implementing additive manufacturing techniques for medical applications, notably for difficult-to-use materials such as PEEK and materials with customized properties.

4.1.8 EUROPEAN SCIENCE COMMUNICATION INSTITUTE (ESCI) GGMBH - ESCI

www.sciencecommunicators.eu

4.1.8.1 DESCRIPTION OF THE LEGAL ENTITY AND ITS MAIN TASKS

The European Science Communication Institute (ESCI) is a not-for-profit organisation that empowers scientists and engineers, helping them to communicate with a non-expert audience to bridge the gap between science, technology and society.

ESCI supports national and international research and innovation initiatives in communicating effectively and leveraging their dissemination potential. In addition to providing up-to-date media training, ESCI has put together a compelling set of dissemination tools specifically with different stakeholders and target audiences in mind. Their multi-skilled team can ensure that the message not only tells, but sells. The ESCI offer includes: optimising web communication, producing state-of-the-art videos, harnessing social media, organising workshops and engaging with end-users to maximise outreach and impact of these initiatives.

As a truly European organisation, ESCI draws on key contributors from across the continent. Their senior science communicators work in France, Belgium, the UK and Germany. The European Science Communication Institute also enjoys close ties with science communication specialists in almost every EU country, which enables it to deliver locally adapted solutions in different languages.

ESCI will lead the communication and dissemination in close cooperation with the exploitation leader. The aim of this activity is to identify the communication needs for the specific audiences and find and use the channels of communication needed to fulfil this task. The ESCI team has experience in web, social media, print, audio-visual and direct 1:1 communication to accomplish a successful mix of communication and dissemination activities. This includes also the tracking of this activity to be able to measure the impact of the communication and dissemination measures in order to adapt it, if necessary.

At ESCI, we believe that the success of a new technology is not only dependent on the technology itself, but also on the way it is communicated and disseminated to the relevant target groups and exploitation channels. For the LASER4SURF project, ESCI see plenty of opportunities to use its creativity and expertise to work together with a strong consortium for a successful communication, dissemination and exploitation of the project.

4.1.8.2 CV OF THE PERSONS

- **Elmar Bartlmae** (male). Mr. Elmar BARTLMAE holds a MSc in Science Communication (1995), Imperial College London and a BSc in Physics with Management (1994) from King's College London. In 2016 he co-founded ESCI and became its Managing Director. Elmar has been involved in different communication and dissemination activities in more than 20 European research projects, including in projects with a focus on complex technologies (Nano-TV, HIPSTER). Initially, Elmar was responsible for video production activities and later got involved all other communication activities, including social media, print and web communication and their dissemination. Elmar is also an award-winning film producer, specialising in science documentaries. In 2016 his co-production The Origami Code on modern mathematics and engineering won the Grand Prix of PARISCIENCE and a Jackson Hole Science Media Award.
- **Alec Walker-Love** (male). Mr. Alec WALKER-LOVE has received a BA in Politics & Modern Languages from the University of Manchester (2003). For the last 15 years, Alec has been managing dissemination and

communication activities of a range of European funded projects including CITyFiED and DIRECTION. Alec is a communication professional, recognised for strategic and original thought. He is passionate about the conversations happening inside and outside organisations and connecting them to the goals of the organisation or project.

- **Naomi Halbach** (female). Ms. Naomi HALBACH holds a BA in Cultural Anthropology/Media Sciences, University of Tübingen (2013) and a BSc in Geosciences, University of Bremen (2016). As science media producer, Naomi has been responsible for a number of videos and multi-media web stories on European research projects, including R2CITIES, MeeFS and HIPSTER. She is also engaged in the public relations activities of the Max-PlanckInstitute of Marine Biology in Bremen.

4.1.8.3 LIST OF RELEVANT PUBLICATIONS

- **Services: CITyFiED** (www.cityfied.eu) is a European project on smart cities, that has been running since 2014. ESCI experts are responsible for large parts of the communication and dissemination activities, including video production, social media and citizen engagement.
- **Services**: **HIPSTER** (www.hipster-project.eu) is a European innovation project on food processing, that has been running since 2015. ESCI experts are managing the communication and dissemination WP, that includes web communication, video production, and targeted communication to stakeholders.
- **Publicastions/Services:** For **R2CITIES** (www.r2cities.eu), a European demonstration project on the renovation of buildings to a high level of energy efficiency, ESCI experts have implemented a new way of multimedia storytelling tool called pageflow. www.science-stories.com/lavatrici .
- **Publications/Services:** For the European Research Project MeeFS (www.meefs-retrofitting.eu), ESCI science media producers have created the final video on the project. https://t.co/9LF33lOhKD
- **Achievement:** The Origami Code is a science documentary produced for 3SAT/France5, which won the most prestigious international prizes in science television, including the Grand Prix AST at the PARISCIENCE Film Festival 2016, and the Science Media Award of Jackson Hole 2016 in the Category “Innovation”. This popular science programme was co-produced by Elmar Bartlmae, co-founder and director or ESCI.

4.1.8.4 LIST OF RELEVANT PREVIOUS PROJECTS

ESCI experts have designed and implemented several Communication and Dissemination plans for EC-funded research and demonstration projects and were also involved in several CSAs. These activities included the design of brochures, posters, production of videos and journalistic articles, their dissemination to science media outlets (mainly for web and TV), communication training for scientists and also the design and implementation of strategies for social media and/or citizen engagement. For the LASER4SURF project, the following five projects had the most similar challenges, in terms of communicating about sophisticated technology projects and ensuring their successful exploitation.

- **Nano-TV** (www.youris.com/nano/nanotv) – Enhancing public awareness on the results of EUropean research actions on nanosciences and nanotechnologies through the profesional use of television media.
- **HIPSTER** (www.hipster-project.eu) – Deployment of High PreSsure and TempeRature food processing for sustainable, safe and nutritious food with fresh-like quality.
- **PIMMS** (http://www.birmingham.ac.uk/research/activity/pimms-itn/index.aspx) – Proton Ionization Molecular Mass Spectrometry (Communication Training only)
- **CommFABNet** (www.commnet.eu) – Comunication of Food, Fishery, Agriculture and Biotechnologies research – a network to support EU-funded research projects
- **DIRECTION** (www.direction-fp7.eu) – Demonstration of very low energy new buildings

4.1.8.5 DESCRIPTION OF ANY SIGNIFICANT INFRASTRUCTURE

No special technical equipment or infrastructure is needed for the proposed activities of ESCI in the LASER4SURF project.

4.1.8.6 OTHER RELEVANT INFORMATION

The European Science Communication Institute was founded in July 2016, and has therefore only a limited official track record of projects or activities to this date. However, the communication experts of ESCI have an impressive experience working in similar projects. As communication and dissemination is a people's business, we have included projects and activities in the sections above that ESCI's founding partners or staff have been responsible for.

4.2 THIRD PARTIES INVOLVED IN THE PROJECT

For all the subcontract activity foreseen in this project the identification of the respective subcontracts will be done on the basis of the best value offer. According to the GA regulation on sub-contracting the criteria used by the partners to assess the "best value" of the different potential subcontractors will be done and proposed on the base of: market tendering, partners' standard practices, commercial agreements already in place. Whatever will be the criteria used by the concerned partner for the respective subcontract, a clear justification of the "best value" will be provided to the Commission in advance of the subcontract finalisation.

CEIT	YES/NO
Does the participant plan to subcontract certain tasks (please note that core tasks of the project should not be sub-contracted) *If yes, please describe and justify the tasks to be subcontracted*	NO
Does the participant envisage that part of its work is performed by linked third parties *If yes, please describe the third party, the link of the participant to the third party, and describe and justify the foreseen tasks to be performed by the third party*	NO
Does the participant envisage the use of contributions in kind provided by third parties (Articles 11 and 12 of the General Model Grant Agreement) *If yes, please describe the third party and their contributions*	NO

LASEA	YES/NO
Does the participant plan to subcontract certain tasks (please note that core tasks of the project should not be sub-contracted) *If yes, please describe and justify the tasks to be subcontracted*	NO
Does the participant envisage that part of its work is performed by linked third parties *If yes, please describe the third party, the link of the participant to the third party, and describe and justify the foreseen tasks to be performed by the third party*	NO
Does the participant envisage the use of contributions in kind provided by third parties (Articles 11 and 12 of the General Model Grant Agreement) *If yes, please describe the third party and their contributions*	NO

MULTITEL	YES/NO
Does the participant plan to subcontract certain tasks (please note that core tasks of the project should not be sub-contracted) *If yes, please describe and justify the tasks to be subcontracted*	NO
Does the participant envisage that part of its work is performed by linked third parties	NO
Does the participant envisage the use of contributions in kind provided by third parties (Articles 11 and 12 of the General Model Grant Agreement)	NO

VISUM	YES/NO
Does the participant plan to subcontract certain tasks (please note that core tasks of the project should not be sub-contracted) *If yes, please describe and justify the tasks to be subcontracted*	NO

Does the participant envisage that part of its work is performed by linked third parties YES

If yes, please describe the third party, the link of the participant to the third party, and describe and justify the foreseen tasks to be performed by the third party

Description of the third party:

IRIS (Innovació i Recerca Industrial i Sostenible, S.L.) will be involved as third party of VISUM.

IRIS is an advanced engineering company that specialises in process monitoring and control, and process optimisation. Its commercial division offers advanced monitoring and control solutions based on: NIR chemical monitoring (handheld device and in-line solutions); hyperspectral imaging solutions for foreign body detection and chemical composition analysis; data management solutions for data processing, decision support, automation and data analysis (big production data). The research division has three core research lines: 1. Optical and spectroscopy technologies (e.g. MEMs spectrophotometers, fluorescence and Raman) for process monitoring for on-line and PAT applications in food, pharmaceuticals and the process industries. 2. Novel processing technologies (ultrasound, plasma, microwave…) for process intensification and non-thermal decontamination. 3. Sustainable processing technologies: waste valorisation strategies, biocompounds and biopackaging for the circular economy

The company was established in 2007 and currently employs 56 staff with a turnover of €3.8M. It is equipped with engineering workshops (3), food testing and wet chemistry laboratory, optics labs (2), and an electronics/telecommunications lab. IRIS is ISO 9001:2008 certified and also is certified under UNE-EN-ISO 166002: Management Systems R&D&i.

Link of the participant to the third party: VISUM is mainly participated by IRIS

Describe and justify the foreseen tasks to be performed by the third party:
The involvement of IRIS will allow to increase the level of excellence of the technical work planned in the project as IRIS staff are familiar with validation of the monitoring devices performance and control systems development.

While VISUM is specialised in optical engineering, IRIS has instruments and specific skills for supporting VISUM in the validation of the developed sensors and in creating control systems.

Key staff at IRIS who will be involved in carrying out this work:

- **Mr. David Montes Monteserin.** (male) BSc in Mechanical Engineering (2011), Polytechnic University of Catalonia (UPC), Barcelona. Prior to joining the Production Processes group at IRIS as Mechanical Design Engineer, he gained wide experience in product design and development, including dosing systems for the chemical industry and both industrial and domestic appliances for the water treatment sector. He has also held positions as Engineering Team Manager, applying renewable energies to industrial-sized drinking water plants among other projects. He is specialised in design for manufacturing and is proficient in SolidWorks, design software which has taught to several companies as a freelancer.
- **MSc. Alejandro Rosales** (male) has a M. Sc. in Physics from the Universidad de La Habana (1986). His background comprises theoretical and technical skills in several scientific and technological areas: optics, lasers, high vacuum, colorimetry, UV-VIS-NIRMIR spectroscopy, chemometrics and design of measurement instruments. He has extensive practical experience in programming, electronics and mechanical CAD design as well as R&D methodology and management. His career has included positions as

	YES/NO

researcher applications engineer, design engineer, business developer and R&D manager. In 2000s, he worked in the FTIR spectroscopy field applied to quality control

in the agro-food (wine) industry, whereby he developed two new measurement instruments for wine analysis. His career as industrial physicist has been focused in applying his wide expertise to provide solutions for industrial problems. Since 2011, he has been the Technical Manager in IRIS, leading a staff of 20 engineers and researchers involved in R&D projects aimed at developing novel high-end solutions for industry. Currently, he is the Science & Technology Manager in IRIS.

- **Dr. Laura Rodríguez Turienzo.** (female) B.S in Food Science & Technology (2008) Masters in Innovation in Safety & Food Technologies (2010), PhD in Biological Sciences (2011), all from University of Santiago de Compostela (USC), Spain. She has participated in a number of R&D regionally and nationally funded projects aimed at developing and validating new bio-based active edible packaging, physico-chemical characterization of novel foods and valorisation of food industry wastes. She is author of several papers in peer-reviewed journals and has participated in different national and international conferences.

The **total estimated budget** of the linked third party is 158.875 €.

Does the participant envisage the use of contributions in kind provided by third parties (Articles 11 and 12 of the General Model Grant Agreement) *If yes, please describe the third party and their contributions*	NO

CICe	YES/NO
Does the participant plan to subcontract certain tasks (please note that core tasks of the project should not be sub-contracted) *If yes, please describe and justify the tasks to be subcontracted*	NO
Does the participant envisage that part of its work is performed by linked third parties *If yes, please describe the third party, the link of the participant to the third party, and describe and justify the foreseen tasks to be performed by the third party*	YES

Description of the third party:

CEGASA Portable Energy (CPE) started working on 28 January 2015 after acquiring several production units of CEGASA Group, a company that definitely disappeared in January 2016. The new company, CPE, was created with the challenge of becoming a global company with a high technological content. For this purpose and from the very beginning, it focused on two areas:

- Production of manganese dioxide with a high level of purity, which allows its application in the manufacture of alkaline batteries
- And the design and manufacture of energy storage systems based on two different technologies zinc-air and lithium-ion.

Furthermore, the area of energy storage systems has reinforced the productive capacity of zinc-air batteries that has made the company the first producer at European level and moreover it has implemented an automated production line of Lithium ion batteries which together with the technological capability, allows CEGASA to be a leading company in this technology in Spain and be at the forefront at European level.

Besides the productive investments made in 2015, it is planned to develop an ambitious investment plan worth 4.5 million euros for the years 2016, 17 and 18. Thanks to this

investment effort CEGASA Portable Energy has developed a range of energy storage systems based on lithium-ion technology that allows to have applications in:

- Mobile systems for buses, vans, motorcycles, etc.
- Stationary systems for residential, UPS, communication stations, industrial applications, etc.
- Large-scale systems as part of smart grids and renewable energy plants.

To achieve these technological developments, CPE in addition to its own resources, usually collaborates with technology centres, universities and other companies

Link of the participant to the third party: CIC and CEGASA form part of the same trust.

Describe and justify the foreseen tasks to be performed by the third party:

The involvement of CEGASA as industrial partner will allow to demonstrate the development of the LASER4SURF project. CIC will work on the validation of the functionalisation as well as on the necessary study to implement the technology for other products like pouch cells.

Key staff at CEGASA who will be involved in carrying out this work:

- Dr. Igor Cantero (R&D Manager, m): Ph.D. in Chemistry (University of the Basque Country, Spain). Since 2000, he has been responsible for the New Technologies R&D Department at CEGASA. Dr. Cantero has authored 8 book chapters and 24 papers published in national and international scientific journals. He also holds 5 patents, has participated in 35 research projects and gave over 30 conference presentations. As responsible of the department, he works in the development of primary lithium, lithium-ion, fuel cell, supercapacitor, rechargeable zinc-air and Li-S or Li-Air technologies.
- Ion Pinedo (Innovation manager, m): Mechanical and Industrial Organisation Engineer by the University of Mondragon (Spain). He starts his activities in CEGASA in 1998 in different alkaline cell production departments and in 2010 he moved to the New Technologies Department. From 2013, he is the responsible of innovation projects. He is involved in the final integration (final assembly + Test and Commissioning) of every energy storage products (lithium batteries and hydrogen fuel cells). He has supervised the entire production chain from the specifications settings to customer's approval such as Abengoa, Repsol, Ingeteam, Jema, Endesa, etc.
- Fernando Perez (Product Homologation Manager, m): Degree in Chemistry by the Autonomous University of Madrid in 1987, also has a Master in Business Administration at the University of Kent. He was responsible for product certification and approval Karey Plastics SA in 1989 until 1992. In 1993 he joined CEGASA being responsible for development of the alkaline battery including product qualification and suppliers. Since 1995, he is a member of EPBA (European Portable Battery Association). In 2010 begins its activities in the Department of New Technologies focusing on the accreditation of suppliers of lithium-ion cells. Currently, he also develops activities of characterization and standardization of the product developed in the department and the tasks associated with LCA.

The **total estimated budget** of the linked third party is 101.125 €.

	YES/NO
Does the participant envisage the use of contributions in kind provided by third parties (Articles 11 and 12 of the General Model Grant Agreement) *If yes, please describe the third party and their contributions*	NO

FAGOR	YES/NO
Does the participant plan to subcontract certain tasks (please note that core tasks of the project should not be sub-contracted) *If yes, please describe and justify the tasks to be subcontracted*	NO
Does the participant envisage that part of its work is performed by linked third parties *If yes, please describe the third party, the link of the participant to the third party, and describe and justify the foreseen tasks to be performed by the third party*	NO
Does the participant envisage the use of contributions in kind provided by third parties (Articles 11 and 12 of the General Model Grant Agreement) *If yes, please describe the third party and their contributions*	NO

RESCOLL	YES/NO
Does the participant plan to subcontract certain tasks (please note that core tasks of the project should not be sub-contracted) The task to be subcontracted are the following: RESCOLL will need to subcontract the biological validation of the surface texturing of implants. Specifically related to evaluating the effect of the structuring on the rate of biofilm formation, as this is one of the key issues (especially) with dental implants and their rejection rate in the body. RESCOLL has the rights accreditations to design and develop Medical Devices (ISO 13485) and to test them (several tests qualified ISO 17 025 and the capacity to perform most of the tests related to the ISO 10993 -17 and -18 (Chemical and physicochemical Biocompatibility)), however, they don´t the required means to develop in vivo and vitro test. The **estimated amount corresponding to the tasks to be subcontracted** is 37.460 €. The test to be perform are the following: ▪ In vitro test. ▪ In vivo test. In vivo test with little animals to test the subcutaneous screening and the implant. ▪ In vivo test. In vivo test with big animals (sheep or dogs).	YES
Does the participant envisage that part of its work is performed by linked third parties *If yes, please describe the third party, the link of the participant to the third party, and describe and justify the foreseen tasks to be performed by the third party*	YES

Description of the third party:

Rescoll Manufacturing will be involved as Third Party of Rescoll.

RESCOLL is a group of companies, concerning this project the important ones are two: RESCOLL and RESCOLL MANUFACTURING (Medical Devices Original Manufacturer). RESCOLL MANUFACTURING belongs 100% to RESCOLL and both RESCOLL and RESCOLL Manufacturing are SMEs (independently and consolidated). RESCOLL Manufacturing is 80% dedicated to manufacture medical devices (ISO 13 485 for manufacturing activities).

	YES/NO

RESCOLL Manufacturing is an independent, privately owned, for-profit SME specialized in the manufacturing of medical devices using traditional methods such as multiaxis milling and turning. Originally known as X-Medical, it was purchased by RESCOLL in 2012 and rechristened RESCOLL Manufacturing.

RESCOLL Manufacturing currently produces a wide variety of implantable materials for different clients in the orthopedic, spinal, and maxillofacial domains, and works with a wide variety of materials. These materials include titanium alloys, chrome-cobalt alloys, surgical stainless steel, and PEEK. In 2016 RESCOLL manufactured more than 80.000 implants. Rescoll Manufacturing offers a wide range of services to control and manufacture medical devices and instruments. We produce a variety of custom implants for orthopaedic, maxillofacial, dental and cardiac fields. Our team's medical knowledge as

well as the understanding of your requirements and expectations allow us to provide a high level service conducted in an ISO 13485 certified environment.

- Spine
- Knee
- Hip
- Maxillofacial
- Cardiac
- Dental
- Ancillary

Rescoll Manufacturing offers a large integrated and qualified technological range (swiss turning machining, 3- to 5-axis machining centers, finishing workshop, laser marking, metrology laboratory…) allowing us to meet your expectations within very short delays. All control and manufacturing stages are conducted in an ISO 13485 certified environment.

- Swiss Turning
- Milling
- Multiaxis Milling / Turning
- Polishing
- Laser Marking
- Cleaning
- Packaging
- Metrology

Rescoll Manufacturing machines all medical grades of implantable and non-implantable medical devices using titanium, stainless steel, steel, cobalt-chrome (phynox…) and polymers including Peek, polyethylene, radel and composite including carbon fiberreinforced PEEK.

- Titanium
- Stainless Steel
- Steel
- Refractory

Link of the participant to the third party: RESCOLL Manufacturing belongs 100% to RESCOLL

Describe and justify the foreseen tasks to be performed by the third party:

	YES/NO

RESCOLL Manufacturing will be one of end-user test case of the LASER4SURF system, particularly on dental implants. It will provide its expertise in medical device design and manufacturing.

Key staff at RESCOLL manufacturing who will be involved in the project:

- **Mr. Fréderic Combarnous** (m), Manager of RESCOLL Manufacturing. Mr. Combarnous manages a team of twelve highly-skilled specialists in medical device manufacturing, and has experience in the organization and execution of multipartner projects through his previous experience at RESCOLL. His expertise in the domain of multipartner projects and medical devices, coupled with the team of technicians under his management, means that RESCOLL Manufacturing is ideally equipped to undertake the LASER4SURF project.

The **total estimated budget** of the linked third party is 25.000 €.

Does the participant envisage the use of contributions in kind provided by third parties (Articles 11 and 12 of the General Model Grant Agreement) *If yes, please describe the third party and their contributions*	NO

ESCI	YES/NO
Does the participant plan to subcontract certain tasks (please note that core tasks of the project should not be sub-contracted) *If yes, please describe and justify the tasks to be subcontracted*	NO
Does the participant envisage that part of its work is performed by linked third parties *If yes, please describe the third party, the link of the participant to the third party, and describe and justify the foreseen tasks to be performed by the third party*	NO
Does the participant envisage the use of contributions in kind provided by third parties (Articles 11 and 12 of the General Model Grant Agreement) *If yes, please describe the third party and their contributions*	NO

5. ETHICS AND SECURITY

5.1 ETHICS

The following ethics issues have been identified:

- Environmental protection and safety:

According to the Ethics Summary Report, during the process of laser tests, the project will cover a representative panel of metal materials (Aluminium, Copper, Titanium and Stainless steel), functionalisation and check thermal conductivity. This can raise the health and safety issue in regard to staff.

In This regard, a specific deliverable D9.1 will be prepared and submitted in M6 providing information related to:

- Further information about the possible harm to the environment and staff caused by the research and state the measures that will be taken to mitigate the risks.
- Confirmation that facilities authorisations have been obtained (e.g. security classification of laboratory), and are kept on file.
- Ensure that appropriate health and safety procedures conforming to relevant local/national guidelines/legislation are followed for staff involved in this project.

5.2 SECURITY

- Activities or results raising security issues NO
- EU-classified information' as background or results NO

6. APPENDIX I - FIGURES FROM WORK PACKAGES DESCRIPTION

FIGURES INCLUDED IN WP2

Task 2.2 Preparation of the experimental laboratory set-up and tests

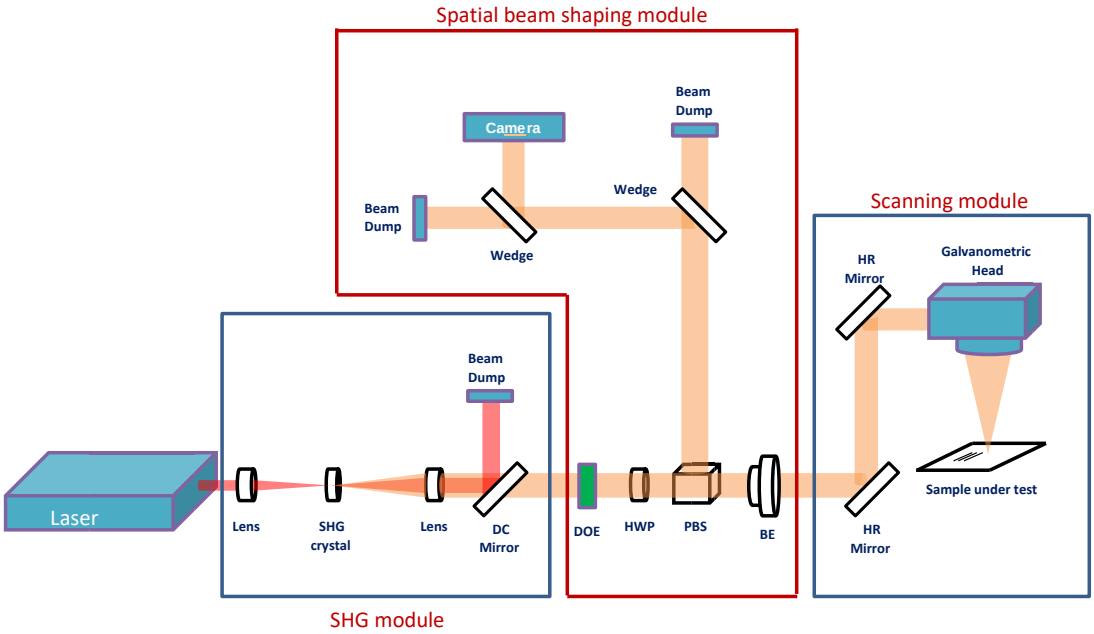
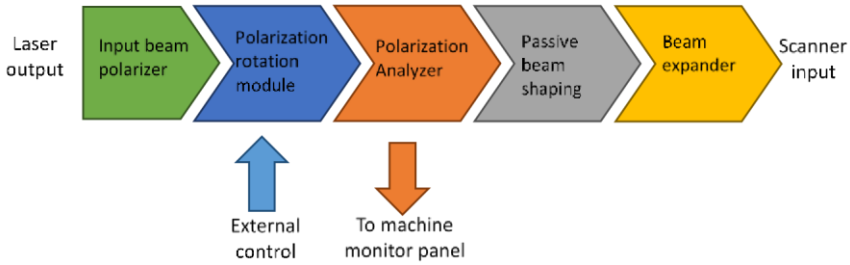


FIGURE 13: CURRENT SET-UP FROM MULTITEL TO BE ADAPTED FOR THE PROJECT

Task 2.4 Implementation and test of the optical modules.



FIGURES INCLUDED IN WP4